

Four-Dimensional Connectedness Loci and Structural Stability for Planar Binary Similarities

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Abstract

We study the connectedness loci of planar binary similarities with independently varying contraction ratios and phases. A canonical normalization gives three four-real-dimensional parameter spaces, according to the orientations of the two similarities. Within this atlas, connectedness is resolved by closed fixed-address contact fibres with exact prefix filtrations and uniform distance bounds. Prescribed symbolic contacts define connected subfamilies; removing their excess identifications gives a relatively G_δ stable set with a constant coding quotient. Exact Cantor-product and parallelogram families provide complete phase diagrams, including a discontinuous radial boundary function. An unequal-ratio triangle family in the opposite–opposite class realizes the universal connected-core bound sharply at every contraction weight. Twelve marked binary zipper charts give smooth two-dimensional subfamilies. A common disk bounds their stable sets; three opposite–opposite families fill this disk exactly and have no stable islands. Finite endpoint-return criteria certify parameter rectangles in five further presentations. Exterior coefficients with moduli greater than one place four selected stable slices in a common three-dimensional projection, with the fourth coordinate retained by colour. Explicit changes of coordinates identify the Koch family underlying the wiggle island. The mathematical constructions are paired with a real-time navigable atlas, reproducible parameter studies and independently verifiable finite disconnection certificates. This brings complex-tree contact families, stable slices and computational exploration into one framework while retaining the distinction between exact classification and finite approximation.

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1 Introduction

A planar self-similar set is obtained by repeatedly replacing a set by smaller transformed copies of itself. In the binary case the attractor is the unique nonempty compact set satisfying

$$K = f_-(K) \cup f_+(K). \quad (1)$$

The connectedness question becomes a problem in four real parameters when the two contraction ratios and two complex phases vary independently. Allowing a reflection in either map adds discrete orientation data. The purpose of the atlas is to treat the resulting parameter loci, and the ways symbolic contacts change along them, as mathematical objects in their own right. Here structural stability is relative to a prescribed parameter family: the full symbolic address-equivalence relation is exactly the relation forced throughout that family. This is a

marked coding notion, not an assertion of ambient openness or of structural stability for a single dynamical map.

The organizing normalization places the translation constants at -1 and 1 . These constants are not generally the fixed points or endpoints of the attractor. After choosing an ordering of the maps, the two complex coefficients give a four-real-dimensional chart for each ordered orientation type. Relabelling identifies the two mixed types, leaving the direct–direct, direct–opposite and opposite–opposite cases. A further reflection quotient can be taken explicitly. This keeps coordinate identifications separate from statements about the connectedness locus itself.

Connectedness is a first-level contact condition, but the form of that contact varies considerably across the atlas. It can be an isolated identification, a more complicated intersection, or an overlap compatible with a connected attractor of positive area. A boundary point of the connectedness locus need not have a unique touching address. The phrase *just-touch frontier* will mean the topological frontier of the connectedness locus, without an additional assumption about contact cardinality or local smoothness.

1.1 Background and the complex-tree programme

Hutchinson’s contraction framework gives existence, uniqueness and approximation of self-similar attractors [1]. Hata’s connectedness theory reduces the binary question to intersection of the first-level pieces [2]. The equal-multiplier locus has a long history beginning with Barnsley–Harrington [3] and Bandt [4]; Calegari–Koch–Walker developed trapping and symbolic methods in their proof of Bandt’s conjecture [5]. Independent complex multipliers in the direct class were studied by Aoki–Fujimura–Taniguchi [21]. The atlas here retains both multipliers and includes opposite similarities, while organizing exact subfamilies by their contacts and coding relations.

The present viewpoint also develops the author’s work on complex trees. Self-contact relations in symmetric tree families were studied in [9]. The master’s thesis [10] and the subsequent framework [11] construct connected self-similar families by prescribing identifications between symbolic addresses and then examine the parameters at which further identifications occur. A ternary fern family is worked out in [12]. The Cornell lecture in 2022 discusses extensions of symmetric families to complex stability regions, including a Koch-curve example [13, 27:11–27:58]. These examples motivate a common description of ambient connectedness loci and algebraic or analytic contact subfamilies within them.

An interactive implementation of the three-family atlas was made publicly available on 24 April 2026 for the author’s UniLab26 workshop at the Universitat de Girona on 25 April [14]. It uses the four coordinates $(\alpha, \gamma, \rho, w)$ retained here. The present account develops this independent programme through exact subfamilies, symbolic contact loci and computational methods.

The author’s doctoral thesis [15] develops the collinear side of this programme through topology, finite capture and restricted polynomial roots. The corresponding results concern specified one-complex-dimensional families [16, 17, 18]. Their extension to independently varying multipliers requires keeping the two address tails and their contraction products separate. The present atlas organizes these broader families by orientation, symbolic contact and zipper structure.

Zippers provide another source of connected families. The endpoint conditions give a continuous parametrization, while injectivity is an additional property [6, 7]. Calegari’s wiggle family is an especially pertinent binary zipper: its embedded-arc locus is disconnected [19]. The particular maps and island theorem occur in the 2022 version; the revision of 7 September

2026 adds an explicit discussion of the two-complex-parameter ambient family and its zipper interpretation. The coordinate comparison below relates these presentations. Calegari–Walker [20] develop boundary dynamics, laminations and external-angle models for the equal-multiplier family. The relation between contact languages in the full atlas and such boundary models is a natural further question.

1.2 Scope and organization

The construction has three complementary parts. First, canonical coordinates and uniform modulus bounds organize the ambient parameter spaces. Exact subfamilies then exhibit dust, critical contact, positive-area overlap and a real antenna in a single parameter diagram. Second, fixed-address fibres resolve connectedness symbolically and describe the additional identifications within a prescribed contact family. This gives a precise connection between the atlas, the complex-tree construction and structural stability. Third, marked zipper charts and finite exclusion methods turn these descriptions into explicit families and reproducible computations.

The exact classifications are given in Corollary 5.2 and Theorem 5.3. Theorem 5.5 proves sharpness of the connected-core bound at every weight in the opposite–opposite class. Theorem 7.1 resolves connectedness by fixed-address fibres, and Proposition 7.3 gives the closed filtration of excess contacts. The marked zipper surfaces are described in Theorem 8.2. Proposition 8.5 and Theorem 8.6 classify three entire stable slices. Proposition 8.7 gives seven finite return criteria. Figures 4–6 display certified inner covers for $DD(1, 0)$, $DD(1, 1)$, $DO(1, 0)$ and $DO(1, 1)$ in exterior coordinates and place them in the ambient atlas. The reflected $DD(0, 1)$ presentation remains part of the symbolic dictionary. These results provide the exact structure used by the computational companion.

The continuous geometry of the attractor, its coding relation and a finite display label carry different information. We keep all three available in the companion explorer. This makes the exact examples useful both as mathematical models and as tests of numerical interpretation. The resulting contribution is a systematic atlas with exact phase families, contact and zipper dictionaries, and a computational method for navigating their parameter loci.

2 Canonical normalization and the atlas coordinates

Put $J_+(z) = z$ and $J_-(z) = \bar{z}$. A contracting planar similarity is $F(z) = c + \lambda J_\kappa(z)$, where $\kappa \in \{+, -\}$ and $0 < |\lambda| < 1$. Its plane orientation is κ . This discrete datum must be distinguished from the endpoint reversal signature of a zipper, introduced below.

Proposition 2.1 (Canonical translation normalization). *Let F_1, F_2 be contracting planar similarities with distinct fixed points. There is a unique complex-affine coordinate change $H(z) = Az + B$, with $A \neq 0$, for which*

$$HF_1H^{-1}(z) = -1 + aJ_{\kappa_1}(z), \quad HF_2H^{-1}(z) = 1 + bJ_{\kappa_2}(z). \quad (2)$$

The coefficients satisfy $0 < |a|, |b| < 1$. Consequently each labelled orientation class has the parameter domain $\mathcal{P} = (\mathbb{D} \setminus \{0\})^2$, of real dimension four. Pairs with a common fixed point have a singleton attractor and are excluded from this normalization.

Proof. The real-affine map $M = (F_1 + F_2)/2$ is a strict contraction, with Lipschitz constant at most $(r_1 + r_2)/2 < 1$. Let c be its unique fixed point and put

$$d = F_2(c) - c = -(F_1(c) - c).$$

If $d = 0$, both similarities fix c , contrary to the hypothesis. Thus $H(z) = (z - c)/d$ is an admissible coordinate change and its conjugated maps have the stated constant terms. A direct coefficient λ is unchanged; an opposite coefficient becomes $\lambda\bar{d}/d$, so contraction ratios and plane orientations are preserved.

Conversely, any such normalization has to send the fixed point of M to the fixed point 0 of the normalized average. It must also send $F_2(c)$ to 1. These two conditions determine H uniquely. If the original maps have a common fixed point, that singleton is their compact invariant set, hence their attractor by uniqueness [1]. Finally, the normalized maps cannot have a common fixed point: such a point would be fixed by their average, hence would be 0, whereas their images of 0 are -1 and 1 . $\square$

We also write $S_- = S_1$ and $S_+ = S_2$ for the two normalized maps. Relabelling identifies the mixed classes $(+, -)$ and $(-, +)$, leaving three unordered plane-orientation types $(+, +), (+, -), (-, -)$. This is a classification of the discrete types, not an assertion that there is only one affine-conjugacy class within each type. In the normalized chart, relabelling followed by $z \mapsto -z$ exchanges (a, b) and the orientation labels; complex conjugation replaces (a, b) by $(\bar{a}, \bar{b})$ and preserves the orientation labels. The normalization removes any further continuous freedom from complex-affine conjugacy. General real-affine coordinate changes, which may include shears, are not part of this equivalence relation.

Write $r_1 = |a|$, $r_2 = |b|$. The coordinates used by the atlas are

$$w = \frac{r_1}{r_1 + r_2}, \quad \rho = \frac{2}{r_1 + r_2}, \quad a = \frac{2w}{\rho} e^{i(\alpha - \gamma)}, \quad b = \frac{2(1 - w)}{\rho} e^{-i(\alpha + \gamma)}. \quad (3)$$

Here $0 < w < 1$ and the strict contraction condition is $\rho > 2 \max\{w, 1 - w\}$. The phase variables belong to a torus:

$$(\alpha, \gamma) \in \mathbb{R}^2 / \Lambda, \quad \Lambda = \langle (\pi, \pi), (\pi, -\pi) \rangle_{\mathbb{Z}}. \quad (4)$$

Indeed, if θ_1, θ_2 are arguments of a, b , then $\alpha = (\theta_1 - \theta_2)/2$ and $\gamma = -(\theta_1 + \theta_2)/2$. One can use $0 \leq \alpha \leq \pi$, with γ modulo 2π , provided the seam $(0, \gamma) \sim (\pi, \gamma + \pi)$ is retained. Endpoints of this rectangle are not independent parameter values.

At fixed w , the remaining parameter domain is three-dimensional. A display of sampled radial transitions therefore provides a way to navigate this section of the atlas. The angular identifications in (4) and the sampled radial intervals are retained with each point. Section 10 specifies the display map and its relation to the underlying parameters.

3 Connectedness, contact and rigorous radial bounds

Fix a plane-orientation type and write $S_1(z) = -1 + aJ_{\kappa_1}(z)$, $S_2(z) = 1 + bJ_{\kappa_2}(z)$. Let K_p , $p = (a, b)$, be the compact attractor. Define

$$\mathcal{M} = \{p \in \mathcal{P} : K_p \text{ is connected}\}, \quad \delta(p) = \text{dist}(S_1 K_p, S_2 K_p).$$

All parameter-space boundaries and closures below are relative to the strict contraction domain unless another ambient space is stated.

Proposition 3.1 (Binary contact criterion). *The attractor K_p is connected if and only if $S_1 K_p \cap S_2 K_p \neq \emptyset$, equivalently $\delta(p) = 0$. The function δ is continuous. Hence $\mathcal{M}$ is relatively closed and its complement is open.*

Proof. Necessity follows because a union of two disjoint nonempty compact sets is disconnected. For sufficiency, put $B = \text{conv}(K_p)$ and $B_{n+1} = S_1 B_n \cup S_2 B_n$, starting at $B_0 = B$. The sets B_n are nested compact sets containing K_p . They are connected by induction: both images of a connected B_n are connected, and their intersection contains $S_1 K_p \cap S_2 K_p$. They converge in Hausdorff distance to K_p , so their intersection is connected. This is the binary instance of Hata's graph criterion [2].

Locally in parameter space, all contraction ratios have a common upper bound $r < 1$. The associated operators on compact sets are uniform contractions in Hausdorff distance, and their maps vary continuously with p . The fixed compact sets therefore vary continuously. Taking continuous images and then the distance between two nonempty compact sets preserves continuity, proving the assertion about δ . $\square$

Proposition 3.2 (Inner and outer bounds). *For each of the three plane-orientation types,*

$$\{p : r_1^2 + r_2^2 \geq 1\} \subseteq \mathcal{M} \subseteq \{p : r_1 + r_2 \geq 1\}. \quad (5)$$

For real coefficients, in every plane-orientation type,

$$\mathcal{M} \cap (-1, 1)^2 = \{(a, b) \in (-1, 1)^2 : |a| + |b| \geq 1\}, \quad (6)$$

with the understood exclusion of zero coefficients from $\mathcal{P}$.

Proof. The diameter D_K of K_p is positive, by distinct fixed points. If the first two pieces intersect, their union has diameter at most the sum of their diameters. Thus $D_K \leq (r_1 + r_2)D_K$, proving the right-hand inclusion.

For the other inclusion suppose the two pieces are disjoint, with distance $d > 0$. Let $V(t)$ be the area of the open t -neighbourhood of K_p . For $0 < t < d/2$ the two corresponding neighbourhoods of the pieces are disjoint, and similarity scaling gives

$$V(t) = r_1^2 V(t/r_1) + r_2^2 V(t/r_2).$$

For a nonempty compact planar set, $V(s) > V(t)$ whenever $s > t > 0$. For example, take a point maximizing a linear coordinate on the compact set; a small open region beyond its t -supporting line lies in the s -neighbourhood but outside the t -neighbourhood. Since $r_j < 1$, the displayed equality is incompatible with $r_1^2 + r_2^2 \geq 1$. This proves the left-hand inclusion, including its equality case. It is the elementary area form of the usual similarity-dimension obstruction to strong separation.

For real a, b , the real line is invariant and $K_p \subset \mathbb{R}$. Let I be its convex hull. The intervals $S_1 I, S_2 I$ have lengths $r_1 |I|, r_2 |I|$, lie in I , and together have convex hull I . Neither interval can contain both endpoints of I , since $r_j < 1$. Thus they cover I exactly when $r_1 + r_2 \geq 1$. In that case I is a compact invariant set and equals K_p . Otherwise their gap separates $S_1 K_p$ from $S_2 K_p$. $\square$

In the atlas coordinates, these bounds read

$$\rho \leq 2\sqrt{w^2 + (1-w)^2} \implies p \in \mathcal{M}, \quad p \in \mathcal{M} \implies \rho \leq 2. \quad (7)$$

Both statements assume the strict contraction condition. Thus the possible transition region lies between these two radii; when $w = 1/2$, they are $\sqrt{2}$ and 2. This localization does not assert that membership is monotone along every radial line.

The exact difference formulation is

$$D_p = aJ_{\kappa_1}(K_p) - bJ_{\kappa_2}(K_p), \quad p \in \mathcal{M} \iff 2 \in D_p. \quad (8)$$

It follows by rearranging the equality of a point in each first-level piece. The product $K_p \times K_p$ is the attractor of the four product maps $S_i \times S_j$ on $\mathbb{R}^4$, and D_p is its image under a real-linear projection. For unequal contraction factors the product maps are generally self-affine. The projection does not automatically provide a finite planar IFS for D_p : its evolving relative coefficients must also be tracked. Plane-orientation parity alone does not close that state space. Any finite graph-directed reduction requires a separate invariance argument.

4 Convex support certificates and rigidity at the outer bound

Write the normalized similarities as

$$f_s(z) = s + L_s z, \quad s \in \{-1, 1\}, \quad L_s = r_s O_s,$$

where O_s is an orthogonal real-linear map and $0 < r_s < 1$. Put $q = \max(r_-, r_+)$, let K be the attractor, and let $P = \text{conv } K$. Inner products below are the Euclidean inner product on $\mathbb{C} \simeq \mathbb{R}^2$, with 1 denoting the real unit vector.

Theorem 4.1 (Convex support certificate). *The support function $h(u) = \sup_{x \in P} \langle u, x \rangle$, $u \in S^1$, is the unique continuous solution of*

$$h(u) = \max_{s \in \{-1, 1\}} (s \langle u, 1 \rangle + r_s h(O_s^\top u)). \quad (9)$$

Define

$$g(u) = r_- h(O_-^\top u) + r_+ h(-O_+^\top u) - 2 \langle u, 1 \rangle. \quad (10)$$

Then $f_-(P) \cap f_+(P) \neq \emptyset$ if and only if $g(u) \geq 0$ for every $u \in S^1$. In particular, one direction for which $g(u) < 0$ certifies that K is disconnected.

Let T denote the operator in (9), and set $h_0 = 0$, $h_n = T^n h_0$. Then

$$0 \leq h_n \leq h_{n+1} \leq h, \quad \|h - h_n\|_\infty \leq E_n := \frac{q^n}{1 - q}. \quad (11)$$

If g_n is obtained by replacing h with h_n in (10), then

$$g_n(u) \leq g(u) \leq g_n(u) + (r_- + r_+) E_n. \quad (12)$$

Consequently, $g_n(u) + (r_- + r_+) E_n < 0$ is a finite disconnection certificate.

Proof. Taking the convex hull of the invariance equation gives $P = \text{conv}(f_-(P) \cup f_+(P))$. The support function of an affine image is obtained by applying the transpose of its linear part to the direction and adding the translation term. This proves (9). The operator T is a contraction on $C(S^1)$ in the supremum norm, with contraction constant at most q , so its fixed point is unique.

The two convex pieces meet precisely when $2 \in L_- P - L_+ P$. Membership in a compact convex set is equivalent to all its supporting half-space inequalities. The support function of $L_- P - L_+ P$ is $r_- h(O_-^\top u) + r_+ h(-O_+^\top u)$, proving the criterion. Actual first-piece contact implies convex first-piece contact.

The operator T preserves pointwise order, and $h_1(u) = |\langle u, 1 \rangle| \geq 0$. Thus its iterates increase to h . Since $\|h_1 - h_0\|_\infty \leq 1$, the contraction estimate gives

$$\|h - h_n\|_\infty \leq \sum_{j=n}^{\infty} \|h_{j+1} - h_j\|_\infty \leq \sum_{j=n}^{\infty} q^j = E_n.$$

Multiplying the pointwise errors by r_- and r_+ proves (12). $\square$

Remark 4.2 (Controlling unsampled directions). Finite directional sampling also requires a bound between samples. The function h_n is the support function of the convex hull of $\{f_u(0) : |u| = n\}$, which lies in the disk of radius $R_n = (1 - q^n)/(1 - q)$. Therefore g_n is Lipschitz in chordal distance on S^1 , with constant at most $L_n = (r_- + r_+)R_n + 2$. If every direction is within chordal distance η of a sampled direction, then $\min_j g_n(u_j) \geq L_n\eta$ proves convex contact. The inequalities must use rigorous numerical enclosures when implemented. Convex contact alone does not certify contact of the fractal pieces; it is an outer approximation to connectedness.

Theorem 4.3 (Rigidity on the outer ceiling). *Suppose $r_- + r_+ = 1$. Then K is connected if and only if both normalized semilinear multipliers are real. In this case K is a real interval and $f_-(K) \cap f_+(K)$ is a singleton. Equivalently, the part of the connectedness locus on $\rho = 2$ consists precisely of these real systems.*

Proof. Assume first that K is connected, and put $D = \text{diam } K > 0$. The level- n cylinder cover has maximum diameter at most $q^n D$ and total diameter

$$\sum_{|u|=n} \text{diam } f_u(K) = D(r_- + r_+)^n = D.$$

Hence $\mathcal{H}^1(K) \leq D$.

We give the equality argument explicitly. Choose a diametral pair $x, y \in K$. If $z \in K$ is not on the segment $[x, y]$, the three strictly positive numbers

$$\begin{aligned} R_x &= \frac{|x - y| + |x - z| - |y - z|}{2}, \\ R_y &= \frac{|x - y| + |y - z| - |x - z|}{2}, \\ R_z &= \frac{|x - z| + |y - z| - |x - y|}{2} \end{aligned}$$

are radii of pairwise disjoint open balls centered at x, y, z . Connectedness implies that the distance from each center takes every value between zero and the corresponding radius on the part of K inside that ball. Since a distance function is 1-Lipschitz,

$$\mathcal{H}^1(K) \geq R_x + R_y + R_z = \frac{|x - y| + |x - z| + |y - z|}{2} > D,$$

a contradiction. Thus $K \subset [x, y]$, and connectedness gives $K = [x, y]$.

It remains to identify the line. The average map $G = (f_- + f_+)/2$ is a strict real-linear contraction fixing zero. Convexity of the segment gives $G(K) \subset K$. Therefore $0 \in K$, and then $-1 = f_-(0)$ and $1 = f_+(0)$ belong to K . Its supporting line is the real line. Each nonzero semilinear map sends a nondegenerate real interval to a real interval, so its complex multiplier is real.

Conversely, for real multipliers the real-slice criterion gives a connected interval when $r_- + r_+ = 1$. The two subintervals cover K , their lengths sum to the length of K , and they are proper subintervals. Their intersection is consequently one point. $\square$

Proposition 4.4 (Area of the first-piece contact). *Let $Q = f_-(K) \cap f_+(K)$ and let m_2 denote planar Lebesgue measure. For every parameter,*

$$m_2(Q) = (r_-^2 + r_+^2 - 1)m_2(K). \quad (13)$$

Every point on the boundary of the connectedness locus, relative to the strict contraction domain, satisfies

$$r_- + r_+ \geq 1, \quad r_-^2 + r_+^2 \leq 1, \quad m_2(Q) = 0.$$

If $r_-^2 + r_+^2 < 1$, one also has $m_2(K) = 0$.

Proof. Inclusion–exclusion and similarity scaling give

$$m_2(K) = r_-^2 m_2(K) + r_+^2 m_2(K) - m_2(Q),$$

which proves the identity. The connectedness locus is relatively closed, so a boundary point belongs to it and satisfies the outer bound. The open set $r_-^2 + r_+^2 > 1$ lies inside the connectedness locus by the universal inner bound. Hence boundary points satisfy the opposite weak inequality. If its left side is strictly less than one, nonnegativity in (13) forces both measures to vanish; at equality the contact measure vanishes directly. $\square$

Zero-area contact can be a nondegenerate segment. The exact families below exhibit this on the full parameter-space boundary.

5 Exact families across the three orientation types

For $0 < t < 1$, define the signed digit set

$$C_t = \left\{ \sum_{n=0}^{\infty} e_n t^n : e_n \in \{-1, 1\} \right\}. \quad (14)$$

It is a Cantor set when $t < 1/2$ and equals

$$I_t = \left[-\frac{1}{1-t}, \frac{1}{1-t} \right]$$

when $t \geq 1/2$. Indeed, the intervals $-1 + tI_t$ and $1 + tI_t$ are disjoint for $t < 1/2$ and cover I_t for $t \geq 1/2$. In the first case the two maps satisfy strong separation, and $\dim_H C_t = \log 2 / (-\log t)$.

5.1 Two complete opposite–opposite families

For $0 < |a| < 1$ compare the two systems

$$f_e(z) = e + a\bar{z}, \quad e \in \{-1, 1\}, \quad (15)$$

$$\tilde{f}_e(z) = e + ea\bar{z}, \quad e \in \{-1, 1\}. \quad (16)$$

Their normalized coefficient pairs are respectively (a, a) and $(-a, a)$.

Theorem 5.1 (Exact opposite–opposite phase diagram). *Both systems have the same attractor and the same labelled geometric first-level pieces. With $t = |a|^2$,*

$$K_a = C_t + aC_t, \quad K_{a,-} = -1 + tC_t + aC_t, \quad K_{a,+} = 1 + tC_t + aC_t. \quad (17)$$

Their coding maps are related by a first-symbol-preserving homeomorphism of the binary address space.

If $a \notin \mathbb{R}$, the following alternatives are exact.

1. For $|a| < 2^{-1/2}$, the attractor is a real-linear image of $C_t \times C_t$, its two pieces are disjoint, and $\dim_H K_a = \log 2 / (-\log |a|)$.

2. For $|a| = 2^{-1/2}$, the attractor is the parallelogram $I_t + aI_t$ and its contact set is the segment aI_t .
3. For $|a| > 2^{-1/2}$, the attractor is the parallelogram $I_t + aI_t$ and its contact is the positive-area parallelogram

$$Q_a = J_t + aI_t, \quad J_t = \left[-\frac{2t-1}{1-t}, \frac{2t-1}{1-t} \right]. \quad (18)$$

For $t \geq 1/2$ their areas are

$$m_2(K_a) = \frac{4|\operatorname{Im} a|}{(1-t)^2}, \quad m_2(Q_a) = \frac{4(2t-1)|\operatorname{Im} a|}{(1-t)^2}. \quad (19)$$

For real a , put $r = |a|$. Then $K_a = C_r$ and

$$Q_a = \emptyset \quad (0 < r < 1/2), \quad Q_a = \left[-\frac{2r-1}{1-r}, \frac{2r-1}{1-r} \right] \quad (1/2 \leq r < 1).$$

Thus the real transition is from a Cantor set to an interval with point contact at $r = 1/2$, and then to an interval with interval contact.

Proof. The real-linear map $A(z) = a\bar{z}$ satisfies $A^2 = t \operatorname{id}$. For the diagonal system, the coding series therefore splits as

$$\pi_{\operatorname{diag}}(e) = \sum_{k \geq 0} e_{2k} t^k + a \sum_{k \geq 0} e_{2k+1} t^k. \quad (20)$$

The even and odd signs vary independently, proving the formula for K_a . Applying either generator and using $C_t = -C_t$ gives the two piece formulas in (17).

For the anti-diagonal system, its length- n linear part is the product of the first n signs times A^n . Consequently

$$\pi_{\operatorname{anti}}(e) = \pi_{\operatorname{diag}}(\Phi(e)), \quad \Phi(e)_n = \prod_{j=0}^n e_j.$$

This map preserves the first symbol and is a homeomorphism: its inverse is $e_0 = d_0$ and $e_n = d_{n-1}d_n$ for $n \geq 1$. It identifies the labelled first pieces as well as the full attractors. No commutation with the shift is asserted.

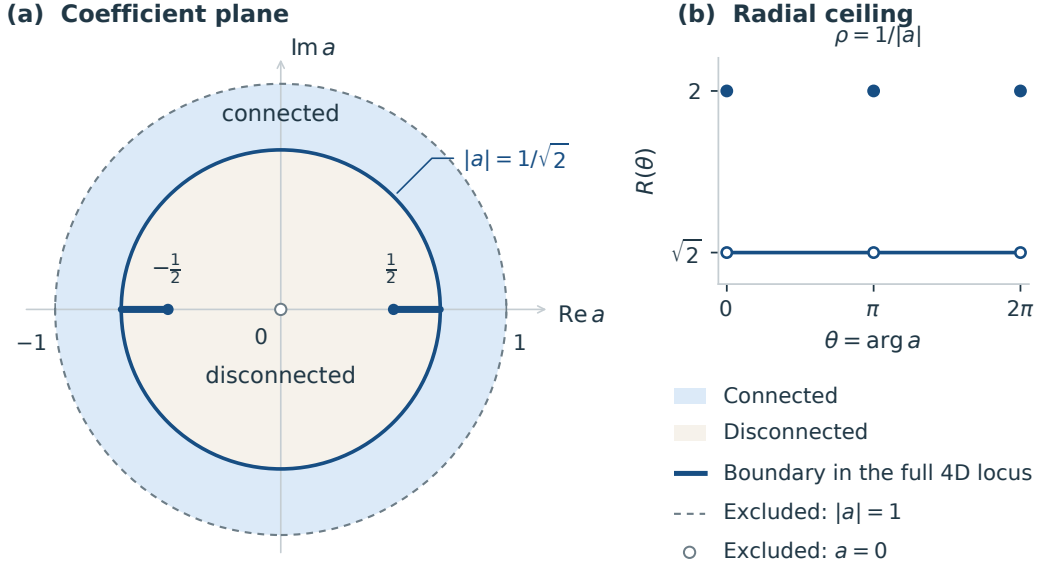
When a is nonreal, $(x, y) \mapsto x + ay$ is an invertible real-linear map. The product representation gives all asserted set types. For $t < 1/2$ the two intervals containing $-1 + tC_t$ and $1 + tC_t$ are separated, and linear independence of $1, a$ makes the two planar pieces disjoint. The product has four strongly separated similarities of ratio t , so its Hausdorff dimension is $\log 4 / (-\log t) = \log 2 / (-\log |a|)$. For $t \geq 1/2$, intersecting the coordinate rectangles gives $J_t \times I_t$. The absolute determinant of the real-linear map is $|\operatorname{Im} a|$, giving (19).

When a is real, the even and odd series interleave into C_r ; the signs of a can be absorbed into independent digit signs. The common first pieces are $-1 + rC_r$ and $1 + rC_r$. Their interval intersection gives the stated formula for Q_a . $\square$

Corollary 5.2 (Exact discontinuous radial frontier). *Let $\mathcal{M}_{OO}$ be the connectedness locus in the full four-real-dimensional opposite-opposite parameter domain. For either embedding $\iota(a) = (a, a)$ or $\iota(a) = (-a, a)$,*

$$\iota^{-1}(\mathcal{M}_{OO}) = \{a : 2^{-1/2} \leq |a| < 1\} \cup \{a \in \mathbb{R} : 1/2 \leq |a| < 1\}, \quad (21)$$

$$\iota^{-1}(\partial \mathcal{M}_{OO}) = \{a : |a| = 2^{-1/2}\} \cup \{a \in \mathbb{R} : 1/2 \leq |a| \leq 2^{-1/2}\}. \quad (22)$$



Both OO slices: $(a, b) = (a, a)$ and $(a, b) = (-a, a)$.

Figure 1: Exact opposite-opposite coefficient slices. The connected region and its intersection with the full four-dimensional boundary follow from Corollary 5.2. The real antennae produce the jumps in the radial ceiling. The unit circle is outside the strict contraction domain.

Writing $a = \rho^{-1}e^{i\theta}$, the connected values on a fixed ray are exactly $1 < \rho \leq R(\theta)$, where

$$R(\theta) = \begin{cases} 2, & \theta \equiv 0 \pmod{\pi}, \\ \sqrt{2}, & \theta \not\equiv 0 \pmod{\pi}. \end{cases} \quad (23)$$

Proof. The connectedness formula is the theorem. Every nonreal point of $|a| = 2^{-1/2}$ is approached along the same ray by disconnected systems with smaller modulus. The real points on the stated segments are approached by nonreal parameters of modulus less than $2^{-1/2}$, also disconnected. The same construction works at the two real endpoints of the circle. All these connected systems are therefore boundary points of the full parameter locus. Conversely, when $|a| > 2^{-1/2}$ the strict inequality $|a|^2 + |a|^2 > 1$ places the parameter in the full connectedness interior. All remaining slice parameters are disconnected, and the disconnectedness set is open. This proves the full-boundary formula, not just a statement about boundary relative to the slice. Substitution of $\rho = 1/|a|$ gives (23). $\square$

The jump is caused by the rank loss of $(x, y) \mapsto x + ay$ on the real axis. It gives an exact example in which a radial frontier function is discontinuous. A continuous interpolated shell can therefore omit connected boundary segments, even on a slice whose radial membership is completely understood.

5.2 A common quarter-turn family in all orientation types

For $0 < r < 1$ consider the following four marked systems:

$$\begin{aligned} \text{DD} : \quad & f_e(z) = e + irz, \\ \text{DO} : \quad & f_-(z) = -1 + irz, \quad f_+(z) = 1 + ir\bar{z}, \\ \text{OO} : \quad & f_e(z) = e + ir\bar{z}, \\ \widetilde{\text{OO}} : \quad & f_e(z) = e + eir\bar{z}. \end{aligned}$$

Theorem 5.3 (Quarter-turn equivalence across orientation types). *All four systems have the same attractor K_r , the same labelled first-level pieces, and the same geometric contact set. With $t = r^2$,*

$$K_r = C_t + irC_t, \quad K_{r,\pm} = \pm 1 + tC_t + irC_t. \quad (24)$$

Their coding maps are related by explicit first-symbol-preserving homeomorphisms of the address space.

For $r < 2^{-1/2}$, the attractor is a Cantor product with $\dim_H K_r = \log 2 / (-\log r)$ and empty first-piece contact. At $r = 2^{-1/2}$,

$$K_r = [-2, 2] + i[-\sqrt{2}, \sqrt{2}], \quad Q_r = i[-\sqrt{2}, \sqrt{2}]. \quad (25)$$

For $r > 2^{-1/2}$, the attractor and its contact are the rectangles

$$K_r = I_t + irI_t, \quad Q_r = J_t + irI_t.$$

In particular, the critical parameter lies on the boundary of the full connectedness locus in each of the three orientation types.

Proof. The two OO cases follow from Theorem 5.1 with $a = ir$. Use the diagonal OO coding map as the reference. In the DD case, splitting even and odd powers of ir gives

$$\pi_{DD}(e) = \sum_{k \geq 0} (-1)^k e_{2k} t^k + ir \sum_{k \geq 0} (-1)^k e_{2k+1} t^k.$$

Thus $\pi_{DD} = \pi_{OO} \circ \Phi_{DD}$, where

$$\Phi_{DD}(e)_{2k} = (-1)^k e_{2k}, \quad \Phi_{DD}(e)_{2k+1} = (-1)^k e_{2k+1}.$$

This coordinatewise involution preserves e_0 .

For the mixed system put $E_k = \prod_{j=0}^k e_{2j}$ and $O_k = \prod_{j=0}^k e_{2j+1}$. A composition of two consecutive linear parts is multiplication by $e_{2j}t$ when acting on a real digit. Grouping these pairs, and then treating the leading odd factor, yields

$$\pi_{DO}(e) = \sum_{k \geq 0} E_k t^k + ir \sum_{k \geq 0} O_k t^k.$$

Equivalently, the coefficient before the digit at position $2k$ is $E_{k-1}t^k$, and the coefficient before the digit at position $2k+1$ is $irO_{k-1}t^k$, with empty products equal to one. These formulas also follow directly by induction on prefix length. Therefore $\pi_{DO} = \pi_{OO} \circ \Phi_{DO}$ with

$$\Phi_{DO}(e)_{2k} = E_k, \quad \Phi_{DO}(e)_{2k+1} = O_k.$$

Its continuous inverse is given by $e_0 = E_0$, $e_1 = O_0$ and $e_{2k} = E_{k-1}E_k$, $e_{2k+1} = O_{k-1}O_k$ for $k \geq 1$. This map also preserves the first symbol.

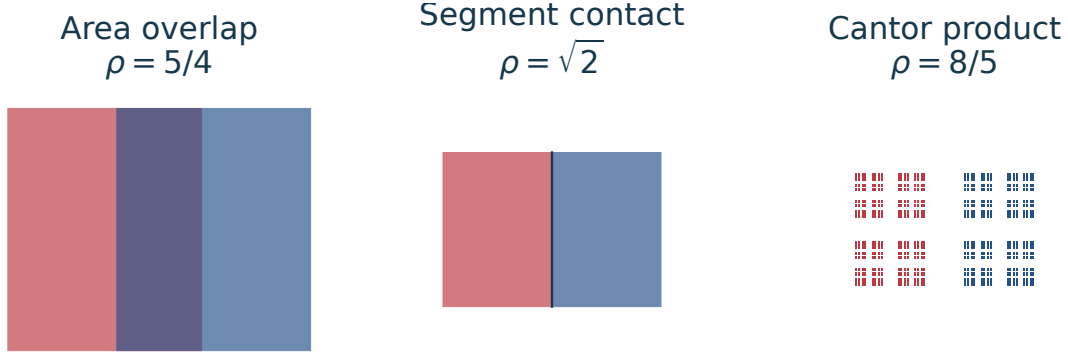


Figure 2: The common quarter-turn family of Theorem 5.3, with $r = 1/\rho$. Red and blue distinguish the first-level pieces in each presentation. The first two panels are exact rectangles and their contact. The last panel shows finite product-cylinder covers of the Cantor attractor, using five terms in each real factor. These are three parameter values from an exact phase classification.

First-symbol preservation transfers the two geometric pieces and their intersection, as well as the attractor itself. The phase classification follows from Theorem 5.1. For every one of the four presentations, approaching the critical value with $r < 2^{-1/2}$ supplies a sequence of disconnected systems in the same orientation type. The critical system is connected, so it lies on the full parameter-space boundary. $\square$

Proposition 5.4 (Address multiplicity on the critical interface). *In any of the four critical presentations, write a contact point as $z = iry$, where $r = 2^{-1/2}$ and $-2 \leq y \leq 2$. If $u = (y + 2)/4$ is an interior dyadic rational, z has four addresses; otherwise it has two addresses, including at $y = \pm 2$. The number of address pairs (ω, η) representing z with $\omega_0 = -1$ and $\eta_0 = 1$ is respectively four or one.*

Proof. In the reference OO coding, zero real coordinate for an address beginning with -1 forces every subsequent even digit to be 1: $-1 + \sum_{k \geq 1} e_{2k} 2^{-k} = 0$ attains the maximum possible tail. An address beginning with 1 similarly forces all subsequent even digits to be -1 . On either side the free odd digits code $y = \sum_{k \geq 0} e_{2k+1} 2^{-k}$. This is the usual binary coding of $u = (y + 2)/4$, which has two expansions exactly at the interior dyadic rationals and one elsewhere. Thus each side supplies two addresses or one address, respectively. Adding counts gives four or two total addresses; multiplying the two side counts gives four or one cross-piece pairs. The first-symbol-preserving recodings transfer these counts to the other presentations. $\square$

This interface has uncountably many geometric contact points, even though all but countably many of them have exactly one address from each first piece. Geometric contact cardinality and address multiplicity are distinct data in an atlas of connected families.

5.3 Unequal-ratio triangles and sharpness at every weight

The preceding common quarter-turn family has equal contraction ratios. Opposite similarities also give an exact boundary family covering every weight $0 < w < 1$.

Theorem 5.5 (An exact triangle on every weighted inner bound). *Let $0 < \theta < \pi/2$, put $c = \cos \theta e^{i\theta}$, and define*

$$g_0(z) = c\bar{z}, \quad g_1(z) = c + (1 - c)\bar{z}.$$

Their attractor is the right triangle $T_\theta = \text{conv}\{0, 1, c\}$. The two pieces have disjoint interiors and meet precisely in the altitude $[c, \cos^2 \theta]$.

The normalized opposite–opposite coefficients are

$$a = -\cos \theta e^{-3i\theta}, \quad b = i \sin \theta e^{-3i\theta}. \quad (26)$$

One atlas representative is

$$\alpha = \frac{\pi}{4}, \quad \gamma = 3\theta - \frac{3\pi}{4}, \quad w = \frac{\cos \theta}{\cos \theta + \sin \theta}, \quad \rho = \frac{2}{\cos \theta + \sin \theta}. \quad (27)$$

Every parameter in this family belongs to the boundary of the full opposite–opposite connectedness locus and to the boundary relative to its fixed-weight parameter space.

As θ ranges over $(0, \pi/2)$, the weight ranges over $(0, 1)$, and the boundary radius is exactly

$$\rho = 2\sqrt{w^2 + (1-w)^2}.$$

Thus the universal connected-core bound is sharp, uniformly over the phases, at every fixed weight in the opposite–opposite class.

Proof. Write $c = x + iy$. Then $x = \cos^2 \theta$, $y = \cos \theta \sin \theta > 0$, and $x^2 + y^2 = x$. Direct substitution gives

$$g_0(0, 1, c) = (0, c, x), \quad g_1(0, 1, c) = (c, 1, x).$$

The identity $\text{Re}(c(\overline{c-1})) = 0$ says that T_θ has its right angle at c . The two image triangles partition it along the altitude from c to x . Uniqueness of the compact invariant set gives the asserted attractor and contact.

For clarity, compute the normalization explicitly. The average map is $G(z) = c/2 + \bar{z}/2$, whose fixed point is $m = x + iy/3$. With $d = g_1(m) - m$ one obtains

$$d = \frac{y}{3}(2y + i(1 - 2x)) = -\frac{iy}{3}e^{2i\theta}, \quad \frac{\bar{d}}{d} = -e^{-4i\theta}.$$

Under $H(z) = (z - m)/d$, the conjugated maps have constant terms -1 and 1 . An opposite multiplier u becomes $u\bar{d}/d$; applying this to c and $1 - c = -i \sin \theta e^{i\theta}$ proves (26). The moduli and arguments give (27), with the usual phase-lattice identifications.

To prove boundary membership, consider, for $0 < t \leq 1$,

$$g_{0,t}(z) = tc\bar{z}, \quad g_{1,t}(z) = c + t(1 - c)\bar{z}.$$

Their images of T_θ remain in T_θ : they shrink the two subtriangles toward 0 and c , respectively. For $t < 1$, the first lies in $\{\text{Re } z \leq tx\}$ and the second in $\{\text{Re } z \geq x\}$. Since $x > 0$, the two pieces are separated. Their attractor, which lies in T_θ , is therefore disconnected.

The normalized coefficient pairs depend continuously on t . Indeed, the mean-map fixed point depends continuously on t , and its normalizing difference is nonzero because $g_{0,t}$ fixes 0 while $g_{1,t}(0) = c \neq 0$. Their contraction ratios are $t \cos \theta$ and $t \sin \theta$, so their weight is independent of t , while their radius is $2/[t(\cos \theta + \sin \theta)]$. The disconnected normalized parameters converge to the connected triangle as $t \uparrow 1$, proving both boundary assertions.

Finally, $w = (1 + \tan \theta)^{-1}$ ranges bijectively over $(0, 1)$, and $w^2 + (1 - w)^2 = (\cos \theta + \sin \theta)^{-2}$. The disconnected sequence at the same weight has radii decreasing to this inner bound. Hence no larger radius can provide a connectedness guarantee uniformly over the phases at that weight. $\square$

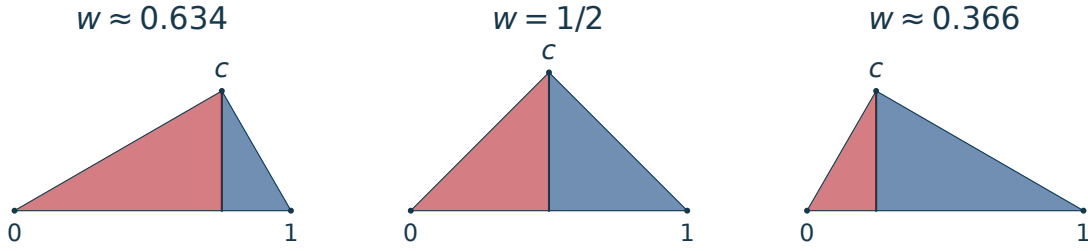


Figure 3: The unequal-ratio opposite triangle family in endpoint coordinates, with $\theta = \pi/6, \pi/4, \pi/3$. The two pieces meet along the altitude. After canonical normalization, every triangle lies at $\rho = 2\sqrt{w^2 + (1-w)^2}$; Theorem 5.5 proves boundary membership and sharpness for every $0 < w < 1$.

At $\theta = \pi/4$, the normalized coefficients are $a = (1+i)/2$, $b = (1-i)/2$, and the normalized triangle and contact are

$$K = \text{conv}\{-3-i, 3-i, 2i\}, \quad Q = [-i, 2i].$$

This supplies an exact opposite–opposite landmark at $(\alpha, \gamma, w, \rho) = (\pi/4, 0, 1/2, \sqrt{2})$.

6 Address relations and stable families

Let $\Sigma_m = \{1, \dots, m\}^{\mathbb{N}}$, and let $\pi_p : \Sigma_m \rightarrow K_p$ be the coding map of a contractive similarity system. We use the following standard implication of the connectedness criterion to specify connected families.

Proposition 6.1 (Connected families from prescribed contacts). *Suppose a parameter family satisfies finitely many coding equalities $\pi_p(i\omega) = \pi_p(j\eta)$ whose first-symbol pairs contain the edges of a connected graph on $\{1, \dots, m\}$. Then every attractor in the family is connected.*

Proof. The prescribed equalities give edges in the first-piece intersection graph. To see the connectedness criterion directly, start with the convex hull of K_p and iterate the union operator, as in Proposition 3.1. At each stage the images are connected and have a connected intersection graph, since they contain the corresponding pieces of K_p . The nested intersection is K_p . $\square$

This separates enforcing connectedness from understanding additional contacts. The complex-tree constructions in [9, 10, 11, 12] motivate treating the latter as a parameter-space problem.

Proposition 6.2 (Regularity of address equations). *For a fixed plane-orientation type, $\pi_p(\omega)$ is real analytic in the normalized coefficients. It is holomorphic in the direct–direct class. If ω is eventually periodic, its value is a rational function of the coefficients and their conjugates; in the direct–direct class it is a holomorphic rational function. Consequently eventually periodic contact equations define relatively real-algebraic sets, and in the direct–direct class complex algebraic sets, after denominators are cleared. A contact equation gives a smooth real surface in the four-dimensional chart at points where its real Jacobian has rank two. No such rank assertion is automatic.*

Proof. Write each map as $t_j + L_j$, where t_j is its translation and L_j is its real-linear similarity part. The coding series is

$$\pi_p(\omega) = t_{\omega_1} + \sum_{n \geq 1} L_{\omega_1} \cdots L_{\omega_n} t_{\omega_{n+1}}.$$

On a compact contraction domain its terms have a geometric uniform bound. For opposite similarities, complexify the coefficients and their conjugates as independent variables. The same series converges normally on a sufficiently small complex neighbourhood where all these variables have modulus less than one. Restriction to the real parameter domain proves real analyticity. In the direct–direct case no conjugate variable occurs, proving holomorphic dependence.

For an address uv^∞ , its value is $S_u(\text{Fix } S_v)$. A direct affine map $c + \nu z$ has fixed point $c/(1 - \nu)$; an opposite map $c + \nu \bar{z}$ has fixed point $(c + \nu \bar{c})/(1 - |\nu|^2)$. Word compositions have polynomial coefficients in the original coefficients and their conjugates, and their fixed-point denominators do not vanish under strict contraction. This proves the rational and algebraic assertions. The final assertion is the real implicit function theorem. $\square$

For nonperiodic addresses the equations are generally analytic, not polynomial. Nor does a contact equation by itself identify a piece of $\partial\mathcal{M}$: it can lie inside $\mathcal{M}$, be a forced identity on a connected slice, or have singular and lower-rank points.

6.1 Explicit eventually periodic contacts in the direct class

Consider $f_-(z) = -1 + az$ and $f_+(z) = 1 + bz$, with $0 < |a|, |b| < 1$. Their fixed points are $p_- = -1/(1 - a)$ and $p_+ = 1/(1 - b)$.

Proposition 6.3 (Cross-contact algebraic families). *For every integer $k \geq 1$,*

$$f_-^k(p_+) = f_+^k(p_-) \iff a^k + b^k = 1. \quad (28)$$

These parameter sets are smooth complex curves within the strict contraction bidisk and every attractor on them is connected. For $k \geq 2$ they lie in the universal connected core $|a|^2 + |b|^2 \geq 1$. For $k > 2$ this inequality is strict, so all such parameters are in the interior of the full connectedness locus. For $k = 2$, equality holds only for real a, b .

Proof. Geometric summation and subtraction give

$$f_-^k(p_+) - f_+^k(p_-) = -\frac{(a + b - 2)(a^k + b^k - 1)}{(1 - a)(1 - b)}.$$

The denominator is nonzero, and $a + b \neq 2$ in the strict contraction bidisk. The two points are coded by $-^k +^\infty$ and $+^k -^\infty$, respectively, so their equality gives first-piece contact. The gradient (ka^{k-1}, kb^{k-1}) is nonzero, proving smoothness of the complex curve.

If $a^k + b^k = 1$, the triangle inequality gives $|a|^k + |b|^k \geq 1$. For $k \geq 2$ and both moduli strictly between zero and one,

$$|a|^2 + |b|^2 \geq |a|^k + |b|^k \geq 1,$$

with strictness in the first inequality for $k > 2$. For $k = 2$, equality throughout requires a^2 and b^2 to be nonnegative real numbers, hence a, b to be real. $\square$

In the atlas coordinates define

$$S_k(\alpha, w) = w^k e^{ik\alpha} + (1 - w)^k e^{-ik\alpha}.$$

Where $S_k \neq 0$, the same contact curves have branches

$$\rho = 2|S_k|^{1/k}, \quad \gamma = \frac{\arg S_k + 2\pi\ell}{k}, \quad \ell = 0, \dots, k-1,$$

retaining only points with $\rho > 2\max(w, 1 - w)$ and imposing the phase-lattice identifications. These are exact connected families with explicit contacts. For $k > 2$ they cannot be frontier sheets; their role is to provide algebraic examples and contact seeds inside the connected region.

6.2 Mirror symmetry and algebraic contact curves

Proposition 6.4 (Mirror-paired contacts). *In the direct-direct chart restrict to $b = \bar{a}$, with $0 < |a| < 1$, and let $\bar{\omega}$ denote the address obtained by exchanging the two symbols of ω . Then*

$$\pi_a(\bar{\omega}) = -\overline{\pi_a(\omega)}.$$

The contact $\pi_a(\omega) = \pi_a(\bar{\omega})$ is therefore equivalent to the single real equation

$$\Re \pi_a(\omega) = 0. \quad (29)$$

Every parameter satisfying this equation belongs to $\mathcal{M}$. If ω is eventually periodic, equation (29) is real algebraic after a nonvanishing denominator is cleared. At a point where the gradient of its left-hand side in $(\Re a, \Im a)$ is nonzero, its zero set is a smooth real-analytic curve; in the eventually periodic case it is a regular branch of a real-algebraic curve.

Proof. The reflection $R(z) = -\bar{z}$ satisfies $RS_1R = S_2$ and $RS_2R = S_1$. Applying these identities to finite words and then passing to the coding limit gives the displayed symmetry. A point is fixed by R exactly when its real part is zero, proving (29). The two paired addresses have opposite first symbols, so their equality puts a point in both first-level pieces and implies connectedness. The algebraicity assertion follows from Proposition 6.2; the gradient condition and the real-analytic implicit function theorem give the local curve. $\square$

This is a precise mechanism for smooth algebraic pieces in mirror-symmetric contact families. It does not make every such piece a connectedness-frontier branch: contact curves may lie in the interior of the ambient connectedness locus. For several simultaneous address equations on the two-real-dimensional mirror slice, one must check their effective number of independent constraints. A single regular scalar equation, or a system of constant rank one in a neighbourhood, gives a curve. Rank one at one point of an arbitrary system alone does not suffice. Rank two can give isolated solutions, while rank zero requires a separate singularity analysis.

In the historical common-translation mirror chart $T_1(z) = 1 + az$, $T_2(z) = 1 + \bar{a}z$, ordinary conjugation exchanges the maps and the corresponding paired contact equation is $\Im \pi_a(\omega) = 0$. For nonreal a , canonical normalization turns that reflection into $R(z) = -\bar{z}$ and preserves the contact condition. At real a the two common-translation maps coincide, so this historical chart degenerates to a singleton; the canonical chart should not be identified with it there.

Definition 6.5 (Forced equivalences and structural stability). For a nonempty parameter family X , define

$$\mathcal{R}_X = \bigcap_{p \in X} \ker \pi_p, \quad \mathcal{S}_X = \{p \in X : \ker \pi_p = \mathcal{R}_X\}.$$

Thus $\mathcal{R}_X$ consists of address identifications forced throughout the family, and $\mathcal{S}_X$ consists of parameters with no additional identifications. We call this structural stability relative to X ; it does not assert dynamical stability or openness in the ambient parameter space.

Proposition 6.6 (Topology on the stable set). *For every $p \in \mathcal{S}_X$, the coding map induces a homeomorphism $\Sigma_m/\mathcal{R}_X \rightarrow K_p$. In particular all attractors in $\mathcal{S}_X$ have the same topology. Neither nonemptiness nor openness of $\mathcal{S}_X$ follows from its definition.*

Proof. Each coding kernel is a closed equivalence relation on compact Σ_m , so their intersection is one also. Its quotient is compact Hausdorff. At a parameter in $\mathcal{S}_X$, the induced continuous map to K_p is bijective, hence a homeomorphism. $\square$

7 Fixed-address contact fibres and stability

The contact criterion can be resolved by fixing one symbolic address and allowing its geometric image to move with the parameter. This retains the address information that is lost when only the connectedness locus is plotted. Throughout this section the orientation type is fixed, $p = (a, b)$ belongs to the strict contraction domain $\mathcal{P}$, and $\Sigma = \{-, +\}^{\mathbb{N}}$. We write $[u]$ for the cylinder with finite prefix u , $\pi_p : \Sigma \rightarrow K_p$ for the coding map, and $K_{p,u} = S_{p,u}(K_p)$. An address is denoted by ω , to distinguish it from the contraction weight w of the atlas.

If $\omega_1 = i$ and $j \neq i$, define the directed contact fibre and prefix locus

$$C_j(\omega) = \{p : \pi_p(\omega) \in S_{p,j}(K_p)\}, \quad (30)$$

$$L_j(u) = \{p : K_{p,u} \cap S_{p,j}(K_p) \neq \emptyset\}, \quad u_1 = i. \quad (31)$$

Both allow every competing address beginning with j . This is different from imposing an equality between two prescribed addresses, and from fixing a spatial point independently of the parameter.

Theorem 7.1 (Symbolic resolution of connectedness). *The sets $C_j(\omega)$ and $L_j(u)$ are relatively closed in $\mathcal{P}$ and*

$$L_j(u) = L_j(u-) \cup L_j(u+), \quad C_j(\omega) = \bigcap_{n \geq 1} L_j(\omega|n). \quad (32)$$

In the binary case,

$$\mathcal{M} = \bigcup_{\omega \in [-]} C_+(\omega). \quad (33)$$

On a compact parameter window $P_0 \subset \mathcal{P}$, put $q = \max_{p \in P_0} \max\{|a|, |b|\} < 1$ and $R = (1 - q)^{-1}$. If

$$\ell_{j,\omega,n}(p) = \text{dist}(K_{p,\omega|n}, S_{p,j}K_p), \quad d_{j,\omega}(p) = \text{dist}(\pi_p(\omega), S_{p,j}K_p),$$

then

$$0 \leq d_{j,\omega}(p) - \ell_{j,\omega,n}(p) \leq 2Rq^n. \quad (34)$$

The functions $\ell_{j,\omega,n}$ increase uniformly to $d_{j,\omega}$.

Proof. The coding map is jointly continuous in the parameter and address: on any compact window its length- n truncation has error at most Rq^n . The collision incidence

$$\{(p, \omega, \eta) : \omega \in [i], \eta \in [j], \pi_p(\omega) = \pi_p(\eta)\}$$

is closed. Projection over the compact symbolic factors proves the closedness assertions, including when the parameter domain is not compact. The identity $K_{p,u} = K_{p,u-} \cup K_{p,u+}$ gives the first formula in (32). The cylinders $K_{p,\omega|n}$ decrease to $\{\pi_p(\omega)\}$; taking their intersections with the fixed compact piece $S_{p,j}K_p$ gives the second formula. The first-level contact criterion gives (33). Finally $\pi_p(\omega)$ belongs to every one of its cylinders, whose diameters are at most $2Rq^n$. Taking distance from the other first-level piece gives (34) and monotonicity. $\square$

The cover (33) is generally not a partition: distinct addresses can determine intersecting fibres. Over a compact parameter window the fibre map is upper semicontinuous. Indeed, a sequence $\omega_n \rightarrow \omega$, $p_n \rightarrow p$, with $p_n \in C_j(\omega_n)$ has a subsequence of competing addresses converging to a competitor for ω at p . This statement concerns the entire fibre; it does not imply that a selected numerical witness varies continuously.

7.1 Inverse orbits and finite outer tests

A word map has the form $S_{p,u}(z) = c_{p,u} + d_{p,u}J_{\kappa_u}(z)$, with $|d_{p,u}|$ the product of its contraction ratios. Set

$$h_{p,u}(\omega) = S_{p,u}^{-1}(\pi_p(\omega)).$$

Then $p \in C_j(\omega)$ if and only if there is an address $\eta \in [j]$ for which $h_{p,\eta|n}(\omega)$ remains bounded. A competing address gives $h_{p,\eta|n}(\omega) = \pi_p(\sigma^n \eta)$. Conversely, similarity scaling gives

$$|\pi_p(\omega) - \pi_p(\eta)| = |d_{p,\eta|n}| |h_{p,\eta|n}(\omega) - \pi_p(\sigma^n \eta)| \longrightarrow 0$$

for every bounded inverse orbit. The argument applies equally to direct and opposite similarities.

Let $q_p = \max\{|a|, |b|\}$, $R_p = (1 - q_p)^{-1}$ and $B_p = \overline{D}(0, R_p)$. Define

$$C_{j,N}(\omega) = \{p : \text{some } u \in \{-, +\}^N, u_1 = j, h_{p,u}(\omega) \in B_p\}.$$

Proposition 7.2 (Exact exhaustion and distance error). *These sets are relatively closed and nested, and*

$$C_j(\omega) = \bigcap_{N \geq 1} C_{j,N}(\omega).$$

If $S_{p,j,N} = \bigcup_{|u|=N, u_1=j} S_{p,u}(B_p)$ and $d_{j,\omega,N}(p) = \text{dist}(\pi_p(\omega), S_{p,j,N})$, then on P_0

$$0 \leq d_{j,\omega}(p) - d_{j,\omega,N}(p) \leq 2Rq^N.$$

Proof. The disk is forward invariant. Thus a surviving word has surviving prefixes, and the finite survivor sets decrease. If every depth has a survivor, König's lemma gives an infinite branch with bounded inverse orbit, which is a competing address by the preceding argument. Conversely every true competitor survives. Finite unions of the continuous closed inequalities $|h_{p,u}(\omega)| \leq R_p$ prove closedness. The nested compact sets $S_{p,j,N}$ contain $S_{p,j}K_p$ and approximate it in Hausdorff distance within $2Rq^N$, proving the distance estimate. $\square$

If an arbitrary address is evaluated using $\widehat{\pi}_{p,L}(\omega) = S_{p,\omega|L}(0)$, its error is at most Rq^L . After inversion along u , the error is at most $Rq^L/|d_{p,u}|$. Consequently the strict inequality

$$\text{dist}(S_{p,u}^{-1}(\widehat{\pi}_{p,L}(\omega)), B_p) > \frac{Rq^L}{|d_{p,u}|} \quad (35)$$

is a sound pruning test when evaluated with validated numerical bounds. Finite survival retains an outer approximation. Membership requires a positive certificate, such as an exact equality with a competing address; when both addresses are eventually periodic their values can be evaluated exactly by the corresponding fixed-point formulas.

7.2 Forced contacts and the stable set of a subfamily

Let $X \subset \mathcal{P}$ be a nonempty subfamily. Put

$$\mathcal{Q}_p = \{(\omega, \eta) \in \Sigma^2 : \omega_1 \neq \eta_1, \pi_p(\omega) = \pi_p(\eta)\}, \quad \mathcal{Q}_X = \bigcap_{p \in X} \mathcal{Q}_p.$$

The relation $\mathcal{Q}_X$ consists of the first-level contacts forced throughout the family. The excess fibre associated with an address is

$$M_X(\omega) = \{p \in X : \text{some } \eta \text{ satisfies } (\omega, \eta) \in \mathcal{Q}_p \setminus \mathcal{Q}_X\}.$$

This is the tip-point construction relative to the prescribed contact structure. It coincides with the full contact fibre when $\mathcal{Q}_X$ is empty. For a zipper surface, $\mathcal{Q}_X$ already includes its endpoint identifications, so $M_X(\omega)$ records additional identifications rather than the contact that makes the whole surface connected.

Proposition 7.3 (Closed approximation of excess instability). *The excess-instability locus $U_X = \bigcup_{\omega \in \Sigma} M_X(\omega)$ is an F_σ subset of X . Every fixed-address excess fibre is F_σ as well. The structural stable set $\mathcal{S}_X = \{p \in X : \ker \pi_p = \mathcal{R}_X\}$ is $X \setminus U_X$ and hence is G_δ in X .*

Proof. If $\mathcal{Q}_X = \emptyset$, the excess locus is the relatively closed connectedness locus restricted to X , and each excess fibre is relatively closed. Otherwise choose any compatible metric on the compact space Σ^2 . For $m \geq 1$ put

$$U_{X,m} = \{p \in X : \exists (\omega, \eta) \in \mathcal{Q}_p, d((\omega, \eta), \mathcal{Q}_X) \geq 1/m\}.$$

The set $\mathcal{Q}_X$ is closed and compact. The collision equation, the different first symbols and the distance constraint are closed conditions; projection over the compact symbolic space makes $U_{X,m}$ relatively closed in X . These sets increase and their union is U_X , since each nonpersistent pair has positive distance from $\mathcal{Q}_X$. Fixing ω gives the same filtration for $M_X(\omega)$.

Finally, every nontrivial equal-address pair has a finite common prefix. Removing that prefix, using injectivity of its word map at every parameter, gives a pair with different first symbols. The original equality is forced throughout X exactly when its reduced equality is. Thus there are no excess first-level pairs exactly when the full coding kernel equals $\mathcal{R}_X$, proving the assertion about $\mathcal{S}_X$. $\square$

This descriptive regularity does not assert openness or nonemptiness. It complements the quotient description of the stable attractors: the connectedness locus is resolved by closed contact fibres, whereas the stable set of a prescribed contact family is obtained by removing its excess identifications.

7.3 Two exact contact curves in the mirror family

The historical complex-tree chart provides explicit examples with one complex parameter. Consider $T_1(z) = 1 + cz$, $T_2(z) = 1 + \bar{c}z$, $0 < |c| < 1$. For the first pair of addresses,

$$\pi_c(12^\infty) - \pi_c(21^\infty) = \frac{(c - \bar{c})(1 - c - \bar{c})}{(1 - c)(1 - \bar{c})}.$$

For the second pair,

$$\pi_c(122(21)^\infty) - \pi_c(211(12)^\infty) = \frac{(c - \bar{c})(1 - 2|c|^2 - |c|^2(c + \bar{c}))}{1 - |c|^2}.$$

Both identities follow by summing the finite prefix and geometric period. Away from the real axis their zero sets are respectively

$$\Re c = \frac{1}{2}, \quad 2|c|^2(1 + \Re c) = 1. \quad (36)$$

The upper branch of the second curve is smooth: with $c = x + iy$ its defining function has y -derivative $4y(1+x) > 0$ when $y > 0$ and $|c| < 1$. Each curve is contained in the corresponding fixed-address contact fibre. Neither equality alone asserts that it exhausts that fibre or lies on the ambient connectedness frontier. At real c the two historical maps coincide; this is the chart degeneration already distinguished from the canonical binary normalization.

8 Zippers: endpoint structure and the binary atlas

The endpoint signature and the plane-orientation type are independent. To keep their roles explicit, use bits $e_j \in \{0, 1\}$ for endpoint reversal and signs $\kappa_j \in \{+, -\}$ for plane orientation.

Proposition 8.1 (General planar similarity zippers). *Let $z_0 = 0, z_1, \dots, z_m = 1$ satisfy $0 < |z_j - z_{j-1}| < 1$, and prescribe e_j, κ_j . The maps*

$$F_j(t) = z_{j-1+e_j} + (-1)^{e_j}(z_j - z_{j-1})J_{\kappa_j}(t) \quad (37)$$

form a zipper. For any partition $0 = t_0 < t_1 < \dots < t_m = 1$, there is a unique continuous map $\zeta : [0, 1] \rightarrow \mathbb{C}$ with $\zeta(0) = 0, \zeta(1) = 1$ such that

$$\zeta \circ h_j = F_j \circ \zeta,$$

where h_j maps $[0, 1]$ affinely onto $[t_{j-1}, t_j]$ and reverses endpoint order exactly when $e_j = 1$. Its image is the IFS attractor. Conversely, every planar similarity zipper with marked ordered endpoints and vertices takes this form after endpoint normalization.

Proof. Formula (37) maps its ordered endpoints to z_{j-1}, z_j with the indicated reversal. Define an operator on the complete space of continuous maps with endpoint values 0, 1 by $(T\zeta) \circ h_j = F_j \circ \zeta$. The endpoint identities ensure that the pieces agree at every partition point and retain the outer endpoints. Its sup-norm contraction ratio is $\max_j |z_j - z_{j-1}| < 1$, so it has a unique fixed map ζ . The compact set $\zeta([0, 1])$ is invariant under the IFS, and therefore is its attractor. Conversely, an endpoint-normalized similarity has coefficient and translation determined by its two endpoint values and its plane orientation, giving the formula. $\square$

This is the standard zipper construction [6]. It covers any finite number of maps, but only the binary case lies in the four-dimensional atlas. An attractor that is a Jordan arc does not necessarily make its given generators a zipper: the two interval maps $0.6t$ and $0.6t + 0.4$, for instance, have attractor $[0, 1]$ with an overlap interval between their pieces.

Let $\tau : \Sigma_m \rightarrow [0, 1]$ be the coding map for h_j . Then $\pi_p = \zeta_p \circ \tau$, so $\ker \tau \subseteq \ker \pi_p$ for every zipper with this partition and endpoint signature. The canonical zipper curve is embedded exactly when these kernels are equal. This distinguishes the canonical embedded-curve locus from the weaker property that the unparameterized attractor happens to be a Jordan arc. It also gives a precise meaning to having only the prescribed endpoint identifications.

For $m = 2$, write $q = z_1$ and $\Omega = \{q : |q| < 1, |1 - q| < 1\}$. Formula (37) becomes

$$F_1(t) = e_1 q + (-1)^{e_1} q J_{\kappa_1}(t), \quad F_2(t) = q + e_2(1 - q) + (-1)^{e_2}(1 - q) J_{\kappa_2}(t). \quad (38)$$

Thus each fixed binary signature and plane-orientation choice has one complex endpoint parameter, independently of the full four-real- dimensional parameter domain of binary systems.

Theorem 8.2 (The binary zipper surfaces). *Fix the plane-orientation type and an endpoint signature $e = (e_1, e_2)$. Within the canonical translation chart, the systems admitting this marked ordered zipper structure form an embedded real-analytic surface parametrized by $q \in \Omega$. It is relatively real algebraic. In the direct-direct class these four surfaces are exactly*

(e_1, e_2)	coefficient equation	
(0, 0)	$a + b = 1$	
(0, 1)	$a - b = 1$	
(1, 0)	$-a + b = 1$	
(1, 1)	$a + b = -1$	(39)

For the other plane-orientation types, the same equations hold in the endpoint coefficient chart defined in the proof, but need not be literal affine planes in the canonical translation coefficients.

Proof. For an arbitrary normalized pair S_1, S_2 , the outer endpoints u, v required by the selected signature are uniquely determined:

e	u	v
$(0, 0)$	$\text{Fix } S_1$	$\text{Fix } S_2$
$(0, 1)$	$\text{Fix } S_1$	$S_2(u)$
$(1, 0)$	$S_1(v)$	$\text{Fix } S_2$
$(1, 1)$	$\text{Fix}(S_1 S_2)$	$S_2(u)$.

They are distinct: equality in any row would give a common fixed point of S_1, S_2 . Set $d = v - u$ and conjugate by $t \mapsto u + dt$. The two maps in these endpoint coordinates have the form

$$\hat{S}_1(t) = AJ_{\kappa_1}(t) - e_1 A, \quad \hat{S}_2(t) = BJ_{\kappa_2}(t) + 1 - (1 - e_2)B, \quad (40)$$

where $A = a$ for $\kappa_1 = +$ and $A = \bar{a}d/d$ for $\kappa_1 = -$, and similarly for B .

For each signature, the transformation $(a, b) \mapsto (A, B)$ is a real-analytic diffeomorphism of the punctured contraction bidisk. To prove surjectivity and the inverse, start with arbitrary (A, B) in the punctured bidisk and the maps (40). The indicated outer endpoint identities hold at 0, 1. These maps cannot have a common fixed point: in each of the four rows the unique fixed points or alternating two-cycle force the distinct outer endpoints. Apply Proposition 2.1 to recover a canonical pair. Taking its signature endpoints returns the original chart, so the two constructions are inverse. All fixed-point operations are real rational by the formulas in Proposition 6.2. The average-map fixed point used in canonical normalization is also real rational: if its linear part is $cz + d\bar{z}$, then $|c| + |d| < 1$, and its fixed point is obtained by an invertible two-by-two real linear system. Thus these inverse changes of chart are real rational, with no vanishing denominator in the contraction domain.

The two inner endpoints in (40) agree exactly when

$$(-1)^{e_1} A + (-1)^{e_2} B = 1.$$

This is a smooth complex affine plane intersected with the bidisk, parametrized by $A = (-1)^{e_1} q$ and $B = (-1)^{e_2} (1 - q)$ for $q \in \Omega$. Pulling back through the real-rational diffeomorphism proves the surface and algebraicity assertions. In the direct-direct class $A = a, B = b$, which gives (39). $\square$

Proposition 8.3 (Embedded binary zippers). *For the binary zipper (38), its canonical curve ζ is injective if and only if $F_1 K \cap F_2 K = \{q\}$.*

Proof. Necessity follows by applying injectivity to the two closed partition intervals. Conversely suppose the intersection is $\{q\}$. Since $q \neq 0, 1$, the point 0 belongs only to the first piece, and 1 only to the second. Let $E_j = \zeta^{-1}(j)$ for $j = 0, 1$. The endpoint equations express each E_j as an affine contracting image, under an appropriate interval map h_i , of one of E_0, E_1 . For signature (0, 0) both sets are self-images; for (0, 1) the first is a self-image and determines the second; for (1, 0) the reverse holds; and for (1, 1) composing the two equations gives a strict contracting self-image for each set. In all cases compactness gives $E_0 = \{0\}$ and $E_1 = \{1\}$, using the prescribed endpoint values.

If two different interval parameters had the same image, remove their longest common interval-cylinder prefix. At the first separation their images lie in the two first-level pieces, hence equal q . Inverting the corresponding injective planar map sends this equality to an

endpoint. At least one preimage parameter is not the partition endpoint, contradicting the uniqueness just proved. Hence ζ is injective. $\square$

All binary zipper surfaces lie in $\mathcal{M}$, regardless of embeddedness. For $q \in (0, 1)$, the endpoint maps act on the real interval as two consecutive subinterval maps; their canonical curve is a homeomorphism. Thus over a whole binary zipper surface the forced kernel is exactly $\ker \tau$, and its structural stable set is the canonical embedded-zipper locus. This does not prove that this locus is connected, or that its components remain distinct after leaving the surface.

8.1 The wiggle slice and the historical tree chart

For plane orientation $(+, +)$ and endpoint signature $(1, 1)$, equation (38) gives Calegari's maps

$$f_q(t) = -qt + q, \quad g_q(t) = (q - 1)t + 1.$$

Their canonical translation coefficients are $a = -q, b = q - 1$. More explicitly, with $Q(q) = q^2 - q + 1$,

$$H_q(t) = \frac{3t - (1 + q)}{Q(q)}$$

conjugates this pair to $-1 - qt$ and $1 + (q - 1)t$. The zeros of Q are $(1 \pm i\sqrt{3})/2$, on the excluded contraction boundary, so this change is valid throughout Ω . Calegari proves that the embedded-curve locus on this slice is disconnected [19]. Every attractor on the slice is connected; the disconnected locus in that result is a parameter space of embeddings, not a family of disconnected attractors.

The common-translation complex-tree maps $1 - qt$ and $1 + (q - 1)t$ are conjugate to f_q, g_q through

$$J_q(t) = \frac{(q^2 - q + 1)t + q - 1}{2q - 1}, \quad q \neq \frac{1}{2}.$$

At $q = 1/2$ the tree maps coincide and their attractor is $\{2/3\}$; the endpoint-normalized wiggle remains $[0, 1]$. This is a genuine chart degeneration. It does not affect the identification at nonreal island parameters, but must be retained when comparing the entire families. Disconnectedness of an embedded-curve locus on this two-dimensional surface does not imply disconnectedness of an embedding locus in the ambient atlas.

8.2 Stable slices of the twelve marked families

For each plane-orientation choice κ and endpoint signature e , let

$$\mathcal{S}_{\kappa,e} = \{q \in \Omega : F_{1,q}K_q \cap F_{2,q}K_q = \{q\}\}.$$

This is the stable set of that marked zipper surface. The parameters in $\Omega \setminus \mathcal{S}_{\kappa,e}$ still give connected attractors; their canonical curves have additional identifications. In particular, these plots resolve structure *within* the connectedness loci. The signs of κ describe orientation in the plane, while the bits of e describe traversal of the two curve pieces.

Proposition 8.4 (A common necessary disk). *For every marked binary planar zipper,*

$$(0, 1) \subset \mathcal{S}_{\kappa,e} \subset \mathbb{D}_{1/2} := \{q : |q - \frac{1}{2}| < \frac{1}{2}\}.$$

Proof. The real interval was established above. If the canonical curve is embedded, its two pieces meet in one point. The planar finite-intersection theorem of Bandt and Rao [22] therefore

gives the open set condition. Choose a bounded nonempty feasible open set O (intersect a feasible open set with a sufficiently large forward-invariant open ball) and write $r_1 = |q|$, $r_2 = |1 - q|$. The disjoint sets $F_1(O), F_2(O) \subset O$ imply $r_1^2 + r_2^2 \leq 1$ by planar area. Equality is impossible for a Jordan arc: in that case $O \setminus \bigcup_{|u|=n} F_u(O)$ has area zero for every n , so the latter union is dense in O . Its distance from K tends uniformly to zero as $n \rightarrow \infty$, which implies $O \subset K$. An arc has empty interior. Hence $|q|^2 + |1 - q|^2 < 1$, equivalent to the displayed disk. $\square$

In the atlas coordinates this necessary disk is $\rho > 2\sqrt{w^2 + (1 - w)^2}$. Stable zipper parameters therefore lie between the guaranteed connected core and the outer bound $\rho \leq 2$; the real interval occupies $\rho = 2$.

Proposition 8.5 (An exact reversing zipper slice). *For $\kappa = (-, -)$ and $e = (0, 0)$, the stable set is exactly $\mathcal{S}_{--,00} = \mathbb{D}_{1/2}$. If $q \notin \mathbb{R}$ lies on its boundary circle, the attractor is the filled triangle $T_q = \text{conv}\{0, q, 1\}$ and the two pieces meet in the segment $[q, |q|^2]$.*

Proof. Conjugation permits $\Im q > 0$. Put $x = \Re q$ and $s = |q|^2$. The maps are $F_1(z) = q\bar{z}$ and $F_2(z) = q + (1 - q)\bar{z}$, and their triangle images are

$$F_1(T_q) = \text{conv}\{0, q, s\}, \quad F_2(T_q) = \text{conv}\{q, 1, 2x - s\}.$$

The strict contraction lens ensures $0 < s < 1$ and $0 < 2x - s < 1$, so both images are contained in T_q . When $q \in \mathbb{D}_{1/2}$, we have $s < x$; the base intervals $[0, s]$ and $[2x - s, 1]$ are disjoint and the triangles meet only at q . Thus $K_q \subset T_q$ and its first-level pieces meet only at q , proving embeddedness. For real $q \in (0, 1)$ the attractor is the interval and embeddedness follows directly. The necessary-disk proposition excludes every remaining parameter. On the nonreal circle $s = x$, the two triangles exactly tile T_q and share the segment from q to s ; uniqueness of the attractor gives the asserted filled triangle. $\square$

8.3 Two further exact stable disks

The opposite–opposite endpoint signatures 01 and 10 have the same complete stable disk as signature 00. In particular, these families supply exact cases with no stable islands.

Theorem 8.6 (Exact stable disks for the reversing signatures 01 and 10). *For the opposite–opposite binary zippers with endpoint signatures $e = (0, 1)$ and $e = (1, 0)$,*

$$\mathcal{S}_{--,01} = \mathcal{S}_{--,10} = \left\{ q : \left| q - \frac{1}{2} \right| < \frac{1}{2} \right\}.$$

Put $s = |q|^2$ and $t = |1 - q|^2$. In the 01 case an invariant triangle on the stable disk is

$$T_{01}(q) = \text{conv} \left\{ 0, 1, \frac{q}{1 - t} \right\};$$

in the 10 case it is

$$T_{10}(q) = \text{conv} \left\{ 0, 1, \frac{q - s}{1 - s} \right\}.$$

For nonreal q on the boundary circle $s + t = 1$, the respective attractors are these filled triangles. The first-level intersection is $[q, 1]$ for signature 01, and $[0, q]$ for signature 10.

Proof. Consider signature 01, whose maps are

$$F_1(z) = q\bar{z}, \quad F_2(z) = 1 - (1 - q)\bar{z}.$$

Complex conjugation permits $\Im q = y > 0$. The real interval $q \in (0, 1)$ gives an embedded interval directly. Define

$$C = \frac{q}{1-t}, \quad A = \frac{s}{1-t}.$$

The contraction lens gives $0 < t < 1$. A direct calculation gives

$$F_2^2(z) = q + tz, \quad F_2(C) = C,$$

and hence

$$\begin{aligned} F_1(T_{01}) &= \text{conv}\{0, q, A\}, \\ F_2(T_{01}) &= \text{conv}\{1, q, C\}. \end{aligned}$$

Inside the disk, $s + t < 1$, so $0 < A < 1$ and $q = (1-t)C$ lies strictly between 0 and C . Every displayed vertex therefore lies in T_{01} , proving invariance.

The line through q and 1 separates these two image triangles. More explicitly, the real-linear functional

$$\ell(z) = \Im(\overline{(1-q)}(z-q))$$

obeys

$$\ell(0) = -y < 0, \quad \ell(A) = (A-1)y < 0, \quad \ell(q) = \ell(1) = 0, \quad \ell(C) = \frac{t}{1-t}y > 0.$$

Thus $F_1(T_{01})$ meets the separating line only at q , while $F_2(T_{01})$ lies on the other closed side. Their intersection is exactly $\{q\}$. The attractor lies in T_{01} , so its first-level pieces also meet exactly at their prescribed join. The binary embedding criterion proves embeddedness throughout the disk. The common necessary-disk theorem excludes every remaining parameter in the contraction lens.

On the nonreal boundary $s + t = 1$, one has $A = 1$. The two image triangles then partition T_{01} along $[q, 1]$ because q lies on the side $[0, C]$. Consequently their union is T_{01} , and uniqueness of the attractor gives the asserted filled triangle and overlap.

Finally, endpoint reflection $z \mapsto 1 - z$ with generator exchange sends signature 01 at $1 - q$ to signature 10 at q . It sends $T_{01}(1 - q)$ to

$$\text{conv}\left\{0, 1, 1 - \frac{1-q}{1-s}\right\} = T_{10}(q)$$

and sends the shared boundary segment $[1 - q, 1]$ to $[0, q]$. □

The finite return tests below also apply to these signatures, but the triangle proof already classifies their complete stable sets.

Finite certificates in seven marked slices. A common endpoint return turns the prescribed contact into a finite verification. Word maps are composed from left to right in symbolic order, so $F_{12} = F_1 \circ F_2$. The following table gives seed words U, V , return words r, s , and the multiplier η of a common similarity $H(z) = q + \eta(z - q)$ satisfying

$$F_{Ur} = H \circ F_U, \quad F_{Vs} = H \circ F_V. \quad (41)$$

Every row holds identically on the contraction lens.

Plane type	e	U	V	r	s	η
DD	01	12	22	1	1	q
DD	10	11	21	2	2	$1 - q$
DD	11	1	2	12	21	$q(1 - q)$
DO	10	11	21	22	22	$ 1 - q ^2$
DO	11	1	2	1212	2121	$ q ^2 1 - q ^2$
OO	01	12	22	11	11	$ q ^2$
OO	10	11	21	22	22	$ 1 - q ^2$

Here DO means that F_1 is direct and F_2 opposite. The identities follow by composition; in the rows with a reversing endpoint return, taking its square produces the displayed positive real multiplier. Since $|\eta| < 1$, H is a strict contraction fixing the join.

Proposition 8.7 (A finite certificate for an embedded zipper). *Fix one row of the table, put $d = |U| = |V|$ and $m = |r| = |s|$, and suppose:*

1. *among the pairs $F_u K \cap F_v K$ with $|u| = |v| = d$, $u_1 = 1$, $v_1 = 2$, every pair other than (U, V) is empty;*
2. *among the pairs $F_{Uu} K \cap F_{Vv} K$ with $|u| = |v| = m$, every pair other than (r, s) is empty.*

Then $q \in \mathcal{S}_{\kappa, e}$. Every required empty intersection can be certified by a finite cover by rigorously disjoint cylinder disks. Conversely, at every parameter in $\mathcal{S}_{\kappa, e}$ both conditions hold and all these disk tests eventually terminate in exact arithmetic. In particular, the seven stable slices in the table are open in Ω .

Proof. The first condition gives $F_1 K \cap F_2 K = J := F_U K \cap F_V K$. The second and (41) give $J = H(J)$. The compact set J contains q and a strict contraction has only one nonempty compact invariant set, namely its fixed point. Hence $J = \{q\}$.

For the finite test, use the invariant disk $B = \overline{D}(1/2, R)$, where

$$R = \max \left\{ \frac{|1 - q|}{2(1 - |q|)}, \frac{|q|}{2(1 - |1 - q|)} \right\}.$$

Both maps send its centre to $q/2$ and $(1 + q)/2$, respectively, regardless of plane orientation or endpoint signature; the formula ensures $F_j(B) \subset B$. Thus $K \subset B$. To establish any required $F_u K \cap F_v K = \emptyset$, recursively replace that pair by its four child pairs, discarding a pair whenever its two cylinder disks are rigorously disjoint. Termination gives a finite cover proving emptiness. Conversely, every such empty compact intersection has positive distance, so this particular pair test eventually terminates in exact arithmetic.

For the converse, an embedded curve has unique endpoint addresses, as in Proposition 8.3. The two addresses of its join are $121^\infty, 221^\infty$ for signature 01; $112^\infty, 212^\infty$ for signature 10; and $1(12)^\infty, 2(21)^\infty$ for signature 11. In each row of the table, (U, V) is their pair of length- d prefixes and (Ur, Vs) the corresponding pair of length- $d + m$ prefixes. Therefore no other tested pair contains the join. Since the two first-level pieces meet only there, all those other pairs are disjoint. The finite tests consequently terminate. Their strict separation inequalities persist locally, which proves openness. $\square$

All conditions in a terminating disk test are strict. When evaluated with outward bounds for a rational rectangle in the q -plane, they certify *every* parameter in that rectangle. A stopped or capped test is unresolved and is not evidence of an additional contact. The DD11 row is the join-renormalization mechanism used for wiggles [19]; the table records the endpoint returns needed for the other marked presentations.

Symmetries and interpretation. Complex conjugation carries each stable slice to itself. Endpoint reflection $z \mapsto 1 - z$, with the two generators exchanged, carries $(\kappa_1, \kappa_2; e_1, e_2; q)$ to $(\kappa_2, \kappa_1; e_2, e_1; 1 - q)$. Thus the DD and OO columns 01 and 10 are reflected companions. In the mixed case that operation interchanges the labels of the direct and opposite maps; it must not be used to identify the displayed DO columns without retaining the relabelling.

8.4 Four stable slices in exterior coordinates

The comparative figures use DD(1, 0), DD(1, 1), DO(1, 0) and DO(1, 1). The first DD signature represents the reflected pair 01, 10; retaining one of them makes the four-family comparison more direct. Each displayed region is a subset of a two-real-dimensional zipper surface in an ambient four-real-dimensional parameter space. The following exterior coordinates place these surfaces in a common geometric display while keeping the fourth coordinate explicit.

For canonical coefficients (a, b) , put

$$c_1 = 1/a, \quad c_2 = 1/b. \quad (42)$$

The strict contraction domain is exactly $|c_1| > 1$, $|c_2| > 1$. These are reciprocal coefficients; if a generator is opposite, its inverse map is $S_j^{-1}(z) = J_{\kappa_j}(c_j(z - t_j))$, with $t_1 = -1, t_2 = 1$. In particular, the conjugation remains part of the inverse map.

It is useful to distinguish this canonical pair from the unsigned endpoint coordinates

$$c = \frac{1}{q}, \quad v = \frac{1}{1 - q} = \frac{c}{c - 1}, \quad (c - 1)(v - 1) = 1. \quad (43)$$

The contraction lens becomes $|c| > 1$, $\Re c > 1/2$, and the common necessary condition for stability becomes the half-plane $\Re c > 1$. Thus the reciprocal chart makes the stable-set bound linear. The planar panels below use this c coordinate. For all four selected families the canonical first coefficient is instead $c_1 = -c$.

Proposition 8.8 (Exterior placement of the four selected surfaces). *Write $q = x + iy$, $s = |q|^2$, and set*

$$N_{10} = 3x - 2y^2 + iy(2x - 1), \quad N_{11} = 1 + q + s(2q - 1).$$

Both quantities are nonzero throughout Ω . For the selected signatures $e = (1, e_2)$, the canonical exterior coefficients are

<i>plane type</i>	c_1	c_2
DD	$-c$	$(-1)^{e_2}v$
DO	$-c$	$(-1)^{e_2}v N_{1e_2}/\overline{N_{1e_2}}$

(44)

Consequently the DO surfaces include a nonconstant phase rotation of the opposite coefficient. Their endpoint equations alone do not give their placement in the canonical coefficient chart.

Proof. Direct coefficients are unchanged by complex-affine conjugacy, giving the DD row. For DO, the first endpoint map is $F_1(z) = q - qz$ and the second has coefficient $B = (-1)^{e_2}(1 - q)$ multiplying $\bar{z}$. Put $D = 3(1 + 2x)$. Solving for the fixed point m of $(F_1 + F_2)/2$ gives

$$m_{10} = \frac{4q + 2\bar{q}}{D}, \quad m_{11} = \frac{1 + 3q + 2s}{D}.$$

The normalizing displacement in Proposition 2.1 is $d = F_2(m) - m = m - F_1(m) = (1 + q)m - q = N_{1e_2}/D$. Hence $a = -q$ and $b = B\bar{d}/d = B\bar{N}/N$, proving the formula after reciprocation. The lens has $0 < x < 1$, so $D > 0$. If $N_{10} = 0$, its imaginary part forces either $y = 0$ or $x = 1/2$. In the first case its real part is $3x > 0$; in the second it is $3/2 - 2y^2 > 0$ because $y^2 < 3/4$. Finally, $\Re N_{11} = 1 - s + x(1 + 2s) > 0$, since $s < 1$ and $x > 0$. $\square$

Projection of the four-dimensional atlas. Use the canonical coefficient phases

$$\theta_1 = \arg a = \alpha - \gamma, \quad \theta_2 = \arg b = -\alpha - \gamma \pmod{2\pi}.$$

Together with

$$w = \frac{|c_2|}{|c_1| + |c_2|}, \quad \rho = \frac{2|c_1||c_2|}{|c_1| + |c_2|}, \quad (45)$$

these give four real coordinates. Define

$$\begin{aligned} X &= (4 + \rho \cos \theta_1) \cos \theta_2, \\ Y &= (4 + \rho \cos \theta_1) \sin \theta_2, \\ Z &= \rho \sin \theta_1. \end{aligned} \quad (46)$$

This map respects the phase lattice (4). At any fixed w , it is injective on the connectedness range $0 < \rho \leq 2$: if $R = \sqrt{X^2 + Y^2}$, then

$$\rho = \sqrt{(R - 4)^2 + Z^2}, \quad \theta_1 = \arg(R - 4 + iZ), \quad \theta_2 = \arg(X + iY).$$

Across different weights the display omits one coordinate. Colour records w , so the projected position together with its numerical weight recovers the exterior pair

$$c_1 = \frac{\rho}{2w} e^{-i\theta_1}, \quad c_2 = \frac{\rho}{2(1-w)} e^{-i\theta_2}.$$

Projected overlaps at distinct weights do not assert intersections in four dimensions. DD and DO remain separate discrete orientation classes and are displayed separately.

On every zipper surface, $w = |q|/(|q| + |1 - q|)$ and $\rho = 2/(|q| + |1 - q|)$. Its stable set lies in

$$2\sqrt{w^2 + (1 - w)^2} < \rho \leq 2. \quad (47)$$

The wireframe $\rho = 2$ in Figure 4 is an outer coordinate reference. It is not a computed connectedness boundary. The central opening is imposed by the toroidal projection.

Certified stable regions and linked curve probes. The coloured surfaces and planar regions are images of closed parameter cells certified by Proposition 8.7. Their union is a finite inner cover of the stable set. The figures show this positive evidence without assigning a membership class to the unpainted part of the plane or the projection. In particular, an apparent gap between cells is not a proved barrier, and the edge of the displayed cover need not be a stable-set boundary. Curved images of cells are drawn by a finite mesh; the underlying certificate is attached to the original q cell. The exterior panels show the finite window $0.96 \leq \Re c \leq 4.65$, $-2.70 \leq \Im c \leq 0.13$. Their lower half corresponds to $\Im q \geq 0$; conjugation gives the other half. The unbounded tail as $q \rightarrow 0$ continues beyond the window.

The same three endpoint parameters link each selected family to three planar curve examples:

$$q_A = \frac{1}{2} + \frac{1}{5}i, \quad q_B = \frac{1}{2} + \frac{2}{5}i, \quad q_C = L = \frac{272727 + 347877i}{800000}. \quad (48)$$

The parameters q_A, q_B are certified stable in all four families. At q_C , DD(1, 1) and DO(1, 1) are certified stable. The DD case is the point in Calegari's disconnected wiggle locus. At the same endpoint parameter, DO(1, 0) has an additional identification, certified by the exact degree calculation described in Appendix A.3. The present certificates do not classify DD(1, 0) at q_C . The three examples therefore compare the geometry of the four marked families at common parameters; they do not assert that all twelve curves are embedded.

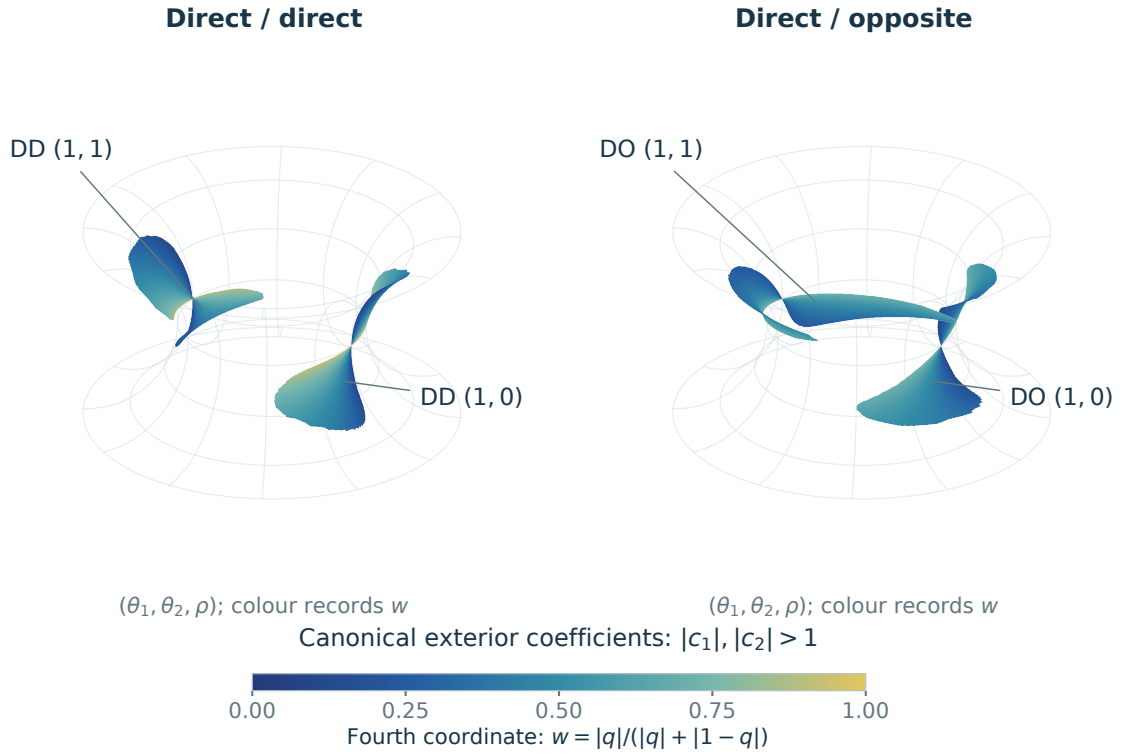


Figure 4: Four selected stable slices placed in the four-dimensional atlas by exterior coefficients $|c_1|, |c_2| > 1$ and projected using (46). DD and DO occupy separate orientation charts. Colour records the fourth coordinate w ; the visible sheets consist of certified stable parameter cells. The wireframe is the outer reference $\rho = 2$. Position alone omits w , and the imposed toroidal opening has no implication for the topology of the connectedness locus. The canonical DO phase rotation is included as in (44).

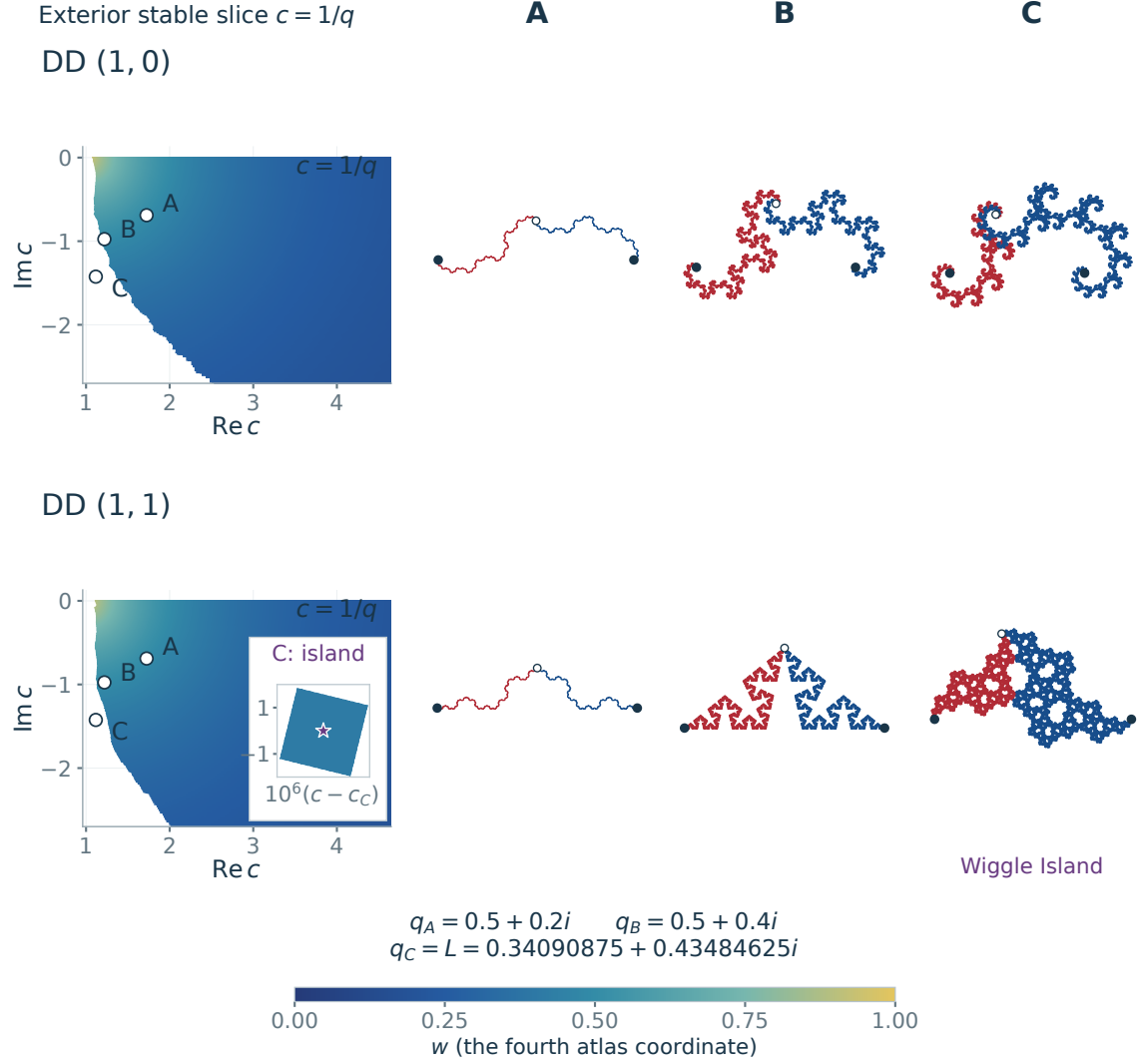


Figure 5: DD(1,0) and DD(1,1): certified stable inner covers in the unsigned endpoint coordinate $c = 1/q$, linked to the three common curve probes (48). The corresponding canonical exterior coordinates are $c_1 = -c$ and $c_2 = (-1)^{e_2}c/(c - 1)$. The DD(0, 1) reflection is represented by DD(1, 0). Red and blue distinguish the first-level curve pieces; they are not membership labels. The tiny certified DD(1, 1) neighbourhood of L is shown separately from the common-scale cover. The curve approximations and their error bounds are recorded in Appendix A.3.

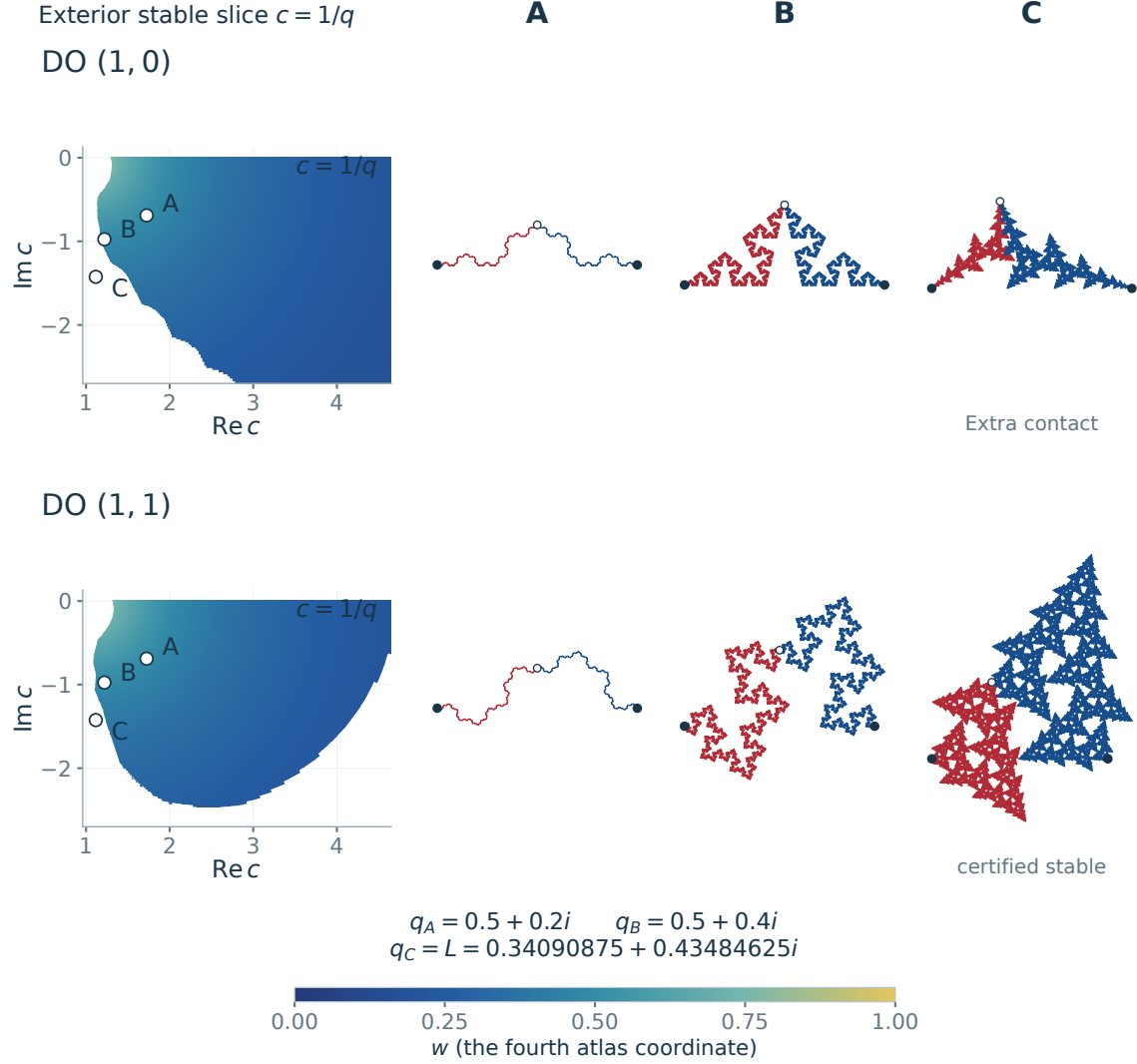


Figure 6: DO(1,0) and DO(1,1) at the same exterior slice coordinates and curve probes as Figure 5. The opposite map contributes the phase factor $N/\bar{N}$ in the canonical exterior coefficient c_2 . The painted slice regions certify stable parameters. The common probe $q_C = L$ produces an additional identification for DO(1,0) and an embedded curve for DO(1,1); these conclusions come from certificates, independently of the finite curve drawings.

9 Finite computation and what its output proves

Let $r = \max\{|a|, |b|\}$ and $R = (1 - r)^{-1}$. The closed disk $B = \overline{D}(0, R)$ satisfies $S_1 B \cup S_2 B \subseteq B$. Put $K_n = \bigcup_{|u|=n} S_u B$. These are nested compact outer approximations of K , with Hausdorff error at most $2Rr^n$.

Proposition 9.1 (Finite disconnection certificates). *The attractor is disconnected if and only if some n satisfies $S_1 K_n \cap S_2 K_n = \emptyset$. Each such assertion reduces to finitely many strict disk-separation inequalities. A parameter that survives these tests to a finite depth has not thereby been proved connected.*

Proof. The sets $S_1 K_n \cap S_2 K_n$ are nested compact sets, and their intersection is $S_1 K \cap S_2 K$. If every one is nonempty, compactness gives a point in their intersection. Otherwise the empty stage certifies disjoint first pieces. Each cylinder image of B is a disk, including for opposite similarities; the finite union intersection is empty exactly when all cross-pairs of disks are disjoint. The binary contact criterion completes the proof. $\square$

Outward-rounded interval or ball arithmetic can turn strict separation inequalities into validated certificates. Finite survival is an outer approximation and should be labelled “not excluded at depth n ”. Positive certificates can instead come from a proved sufficient region such as (5), a verified zipper relation, or exact address contacts forming a connected first-piece graph.

Lemma 9.2 (Persistence of a positive contact gap). *Fix the plane orientations and let the two coefficient pairs λ and μ have contraction ratios at most $\bar{q} < 1$. Put $\eta = \max_j |\lambda_j - \mu_j|$. For the corresponding attractors and first-piece distances,*

$$d_H(K_\lambda, K_\mu) \leq \frac{\eta}{(1 - \bar{q})^2}, \quad |\delta(\lambda) - \delta(\mu)| \leq \frac{2\eta}{(1 - \bar{q})^2}.$$

If $\delta(\lambda) \geq d_0 > 0$ and $\eta \leq d_0(1 - \bar{q})^2/4$, then K_μ is disconnected with first-piece gap at least $d_0/2$.

Proof. Both attractors lie in the disk of radius $R = (1 - \bar{q})^{-1}$. Their Hutchinson operators differ by at most ηR on every compact subset of this disk. The contraction estimate therefore gives

$$d_H(K_\lambda, K_\mu) \leq \bar{q} d_H(K_\lambda, K_\mu) + \eta R,$$

which proves the first bound. Either corresponding first-level piece is displaced by at most $\bar{q} d_H(K_\lambda, K_\mu) + \eta R \leq \eta/(1 - \bar{q})^2$. The distance between two compact sets changes by at most the sum of their Hausdorff displacements, proving the second bound and its consequence. $\square$

For a rational centre with a rational verified gap bound $d_0 > 0$, let $q < 1$ be a rational outward bound for both multiplier moduli and put $\bar{q} = (1 + q)/2$. The radius

$$\eta_0 = \min \left\{ \frac{1 - q}{2}, \frac{d_0(1 - \bar{q})^2}{4}, \frac{1}{2} \min_j \max(|\Re \lambda_j|, |\Im \lambda_j|) \right\} \quad (49)$$

defines a closed product of complex disks $|\mu_j - \lambda_j| \leq \eta_0$ lying in the punctured contraction bidisk. Every parameter in this product has gap at least $d_0/2$. Its interior is an explicit open disconnected region in four real dimensions.

An address chosen by a finite search is a witness selected by that algorithm and its tie-breaking rule. A colour change between such addresses does not, without further proof,

establish a topological stratum, an entropy invariant or a smooth frontier branch. Boundary meshes should therefore record the depth, enclosure bound, decision status and witness-selection rule. Radial scans must retain all detected transition intervals unless a one-transition theorem has been established for the region being rendered.

10 From symbolic contact to a navigable atlas

The companion *Connectedness Atlas* implements a common passage from a parameter to its maps, symbolic cylinders, contact data and planar geometry. The parameter inspector and exported specimen records retain the four coordinates $(\alpha, \gamma, \rho, w)$ and the orientation pair. Thus a point chosen in a three-dimensional display can be returned to an explicit planar system. The implementation and mathematical conventions are available at <https://complextrees.com/4D/>; the source snapshot discussed here was inspected on 18 September 2026.

10.1 Paired words and their enclosures

Use $\lambda_- = a$, $\lambda_+ = b$ and the signs κ_j of Section 2. An accumulated cylinder map is stored as

$$F_u(z) = t_u + L_u J_{\kappa_u} z.$$

Appending a letter j on the right gives the recurrence

$$t_{uj} = t_u + L_u J_{\kappa_u}(t_j), \quad L_{uj} = L_u J_{\kappa_u}(\lambda_j), \quad \kappa_{uj} = \kappa_u \kappa_j, \quad (50)$$

where signs are multiplied. The implementation stores them as parity bits. This outermost-first convention makes the first address symbol identify an actual first-level piece, including for orientation-reversing maps.

The marked difference for the normalized system is

$$D_p = \lambda_- J_{\kappa_-}(K_p) - \lambda_+ J_{\kappa_+}(K_p), \quad K_p \text{ is connected} \iff 2 \in D_p.$$

An unequal-multiplier or reversing system is evaluated through two accumulated maps and two parity bits. On the common direct-multiplier slice this reduces to the usual homogeneous difference construction. Equal contraction ratios alone do not make that reduction valid.

Let $q = \max(|\lambda_-|, |\lambda_+|) < 1$ and $R = (1 - q)^{-1}$. The invariant disk $\overline{B}(0, R)$ supplies each word with the enclosure

$$F_u(K_p) \subseteq \overline{B}(t_u, R|L_u|).$$

A cross-prefix pair (u, v) can therefore be discarded when $|t_u - t_v| > R(|L_u| + |L_v|)$. The implementation develops the remaining pairs in four children. Complete exhaustion gives finite separation; otherwise the output retains a surviving frontier or the reason that a work limit stopped the search. The browser uses this procedure to locate parameters worth refining. For exact rational coefficients, a separate implementation replaces modulus estimates by rational outward bounds and compares squared distances exactly.

The resulting proof object is a finite prefix antichain covering every infinite cross-prefix address pair. Its independent checker reconstructs word maps with rational real 2×2 matrices, verifies cover completeness and checks every strict separation inequality. A successful record also supplies a positive contact gap and an explicit product of complex parameter disks on which that gap remains positive. This provides a constructive route from an atlas sample to an open disconnected region in four real dimensions. It does not require the visualization to decide membership.

10.2 Radial surveys and three-dimensional navigation

At a fixed weight and orientation, the survey samples the angular variables and searches the radial interval

$$2\sqrt{w^2 + (1 - w)^2} \leq \rho \leq 2.$$

The inner endpoint comes from the closed square-sum connectedness criterion; the outer endpoint comes from the sum-of-ratios obstruction. The current three-orientation explorer samples an entire radial interval, brackets each detected survival–exclusion or exclusion–survival transition, and refines those brackets. It records re-entry bands and unresolved searches. The visible surface uses the outermost detected finite-survival sheet. A narrow band may fall between samples, so the recorded transition list remains part of the data.

The detailed direct–direct laboratory at <https://complextrees.com/4D/lab/> contains $13 \times 41 \times 81 = 43\,173$ sampled rays: thirteen weights from 0.35 to 0.65, with 41 values of α and 81 values of γ . It retains two search depths, the first and outer exits, band counts and cross-depth discrepancies. A rectangular chart displays the original stored samples, while angular cosine interpolation and a cubic spline in weight supply the smooth three-dimensional display. Selecting or exporting a stored cell retains its original data. The display mesh and the radial survey therefore have separate resolutions.

Paired letters are encoded by four symbols. The atlas can colour a parameter by a selected surviving finite word pair or compare complete retained prefix data. Such fields organize numerical experiments and expose changes that can be examined through the address-equivalence relation. A constant finite label is not used as the definition of stability. Likewise, a toroidal embedding is a coordinate display; its imposed central hole does not classify the topology of the connectedness locus.

10.3 Exact families as navigable subsets

The family registry places exact parameterized systems in the same coordinates as the free atlas. In particular, it implements the twelve marked binary zipper presentations obtained from three unordered planar orientation types and four traversal signatures. Each uses the strict lens $|z| < 1$, $|1 - z| < 1$, evaluates its endpoint maps, and normalizes the resulting pair. The traversal bits and the planar parity remain separate fields. These are twelve marked presentations; no assertion that their images are pairwise disjoint or that they classify every connected binary attractor is needed. The comparative figures select $DD(1, 0)$, $DD(1, 1)$, $DO(1, 0)$ and $DO(1, 1)$. Their coefficient-phase projection is defined in Section 8.4; it preserves the canonical normalization of the registry, including the phase rotation of the opposite map. It is a separate figure construction from the earlier (α, γ) display of the six-specimen survey below.

The same normalization accepts the binary complex-tree families defined by explicit eventually periodic address relations. Their records carry the raw tree maps, the contact words, the cleared relation and the resulting normalized parameter. The reader can therefore move between the original tree alphabet, an algebraic contact family, a marked zipper when available, and the ambient four-dimensional chart without changing conventions silently.

10.4 Finite geometry, hulls and empirical difference images

The planar renderer enumerates all words at the reported depth and carries their first-level piece and parity. It computes in binary64 and draws binary32 vertex buffers using a worker and WebGL 2, with a Canvas fallback. Geometry budgets choose the actual depth; the export

Sample	α/π	γ/π	ρ	Rq^{22}	$2qRq^{22}$
A	$\frac{107}{150}$	$\frac{62}{75}$	1.4677798748	6.76×10^{-4}	9.21×10^{-4}
B	$\frac{109}{150}$	$\frac{83}{75}$	1.5546487570	1.70×10^{-4}	2.19×10^{-4}
C	$\frac{17}{50}$	$\frac{44}{75}$	1.6338859797	5.25×10^{-5}	6.43×10^{-5}
D	$\frac{67}{150}$	$\frac{39}{25}$	1.5326993465	2.39×10^{-4}	3.12×10^{-4}
E	$\frac{1}{50}$	$\frac{34}{75}$	1.4743398428	6.07×10^{-4}	8.24×10^{-4}
F	$\frac{1}{50}$	$\frac{16}{75}$	1.5181920528	3.00×10^{-4}	3.96×10^{-4}

Table 1: Parameters of the six equal-contraction DD specimens ($w = 1/2$), with $q = 1/\rho$. Radial values are rounded here; the full numerical record is retained in the source. The last columns are analytic tail estimates in exact arithmetic, excluding floating-point and density-grid errors.

records that depth and the number of prefixes drawn. Camera movement and a finer raster do not change the underlying word set.

In exact arithmetic, with $Z_N = \{F_u(0) : |u| = N\}$,

$$d_H(K_p, Z_N) \leq Rq^N, \quad d_H(D_p, D_{p,N}^{\text{ctr}}) \leq (|\lambda_-| + |\lambda_+|)Rq^N, \quad (51)$$

where $D_{p,N}^{\text{ctr}}$ uses the two transformed copies of Z_N . Convexification is nonexpansive for Hausdorff distance, so these bounds also compare the filled finite convex hulls with the corresponding limiting convex hulls. The drawn outlines are computed from finite floating-point centres; the analytic tail bound does not by itself enclose every floating-point error.

The six-sample study at <https://complextrees.com/4D/atlas/> uses $N = 22$, hence 2^{22} addressed centres per attractor. Its difference images retain word multiplicities by taking a histogram cross-correlation of the two transformed word clouds. This represents 2^{44} ordered pairs without enumerating every pair explicitly. The quantity displayed is the probability per unit area for independent uniform finite words on the specified lattice. A shared logarithmic opacity scale permits comparison across specimens. For nearest-lattice rounding with spacing h , rounding the two clouds before subtraction adds at most $\sqrt{2}h$ of geometric displacement, in addition to the finite-tail and arithmetic errors. This finite distribution is not assumed to be a density of a limiting measure.

10.5 Six linked parameter specimens

Figures 7 and 8 provide a linked view of the parameter atlas and the planar sets it describes. The six samples share $w = 1/2$ and direct orientations, but generally have different complex multipliers. Table 1 records the phase representatives and radial values used by the figures. In this display the angular embedding is

$$(\alpha, \gamma, \rho) \mapsto ((4 + \rho \cos \alpha) \cos \gamma, \rho \sin \alpha, (4 + \rho \cos \alpha) \sin \gamma).$$

It is a visualization of a phase-chart representative, with the seam metadata retained; it is not asserted to be a globally injective embedding of the phase-lattice quotient. A specimen links that display position to two planar objects: K_p , with its first-level decomposition, and D_p , with the target point 2. In this way the same parameter can be examined through ambient position, geometry and finite symbolic contact information.

10.6 Reproducible states and figures

The mathematical record consists of coefficients, orientations, word convention, sampled grids, budgets, retained evidence and arithmetic. The figure record adds the camera, projection,

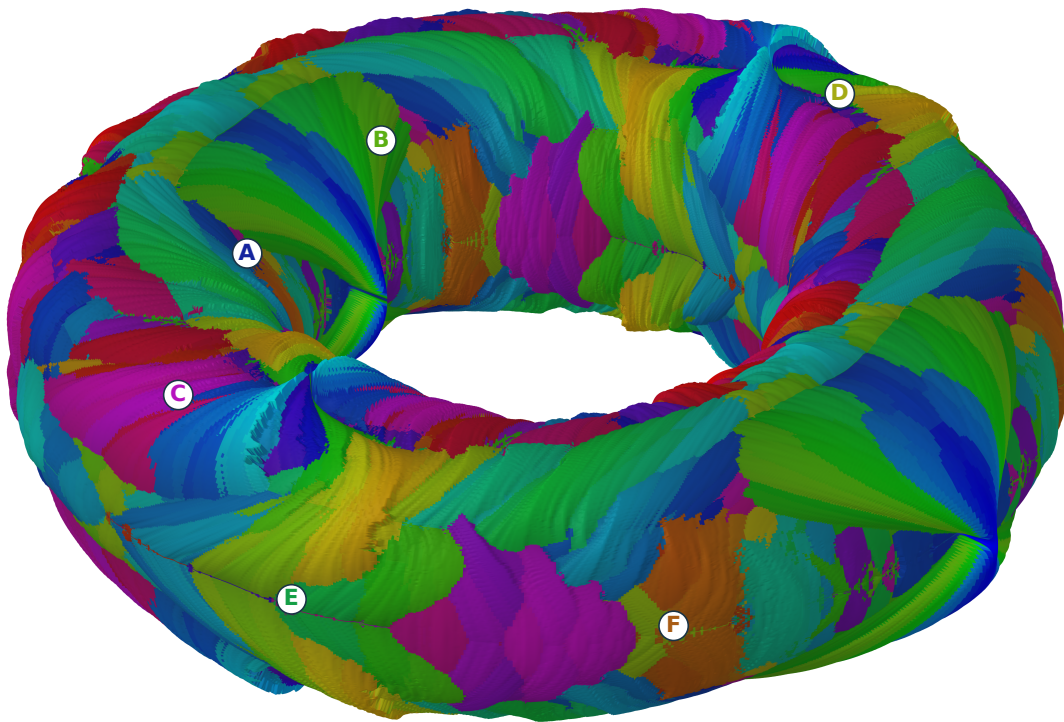


Figure 7: Equal-contraction DD parameter atlas with specimens A–F. The angular view displays the finite numerical study underlying the companion explorer. Colours record selected finite pair-address codes. The central hole belongs to the coordinate display. Parameters are given in Table 1, with planar sets in Figure 8.

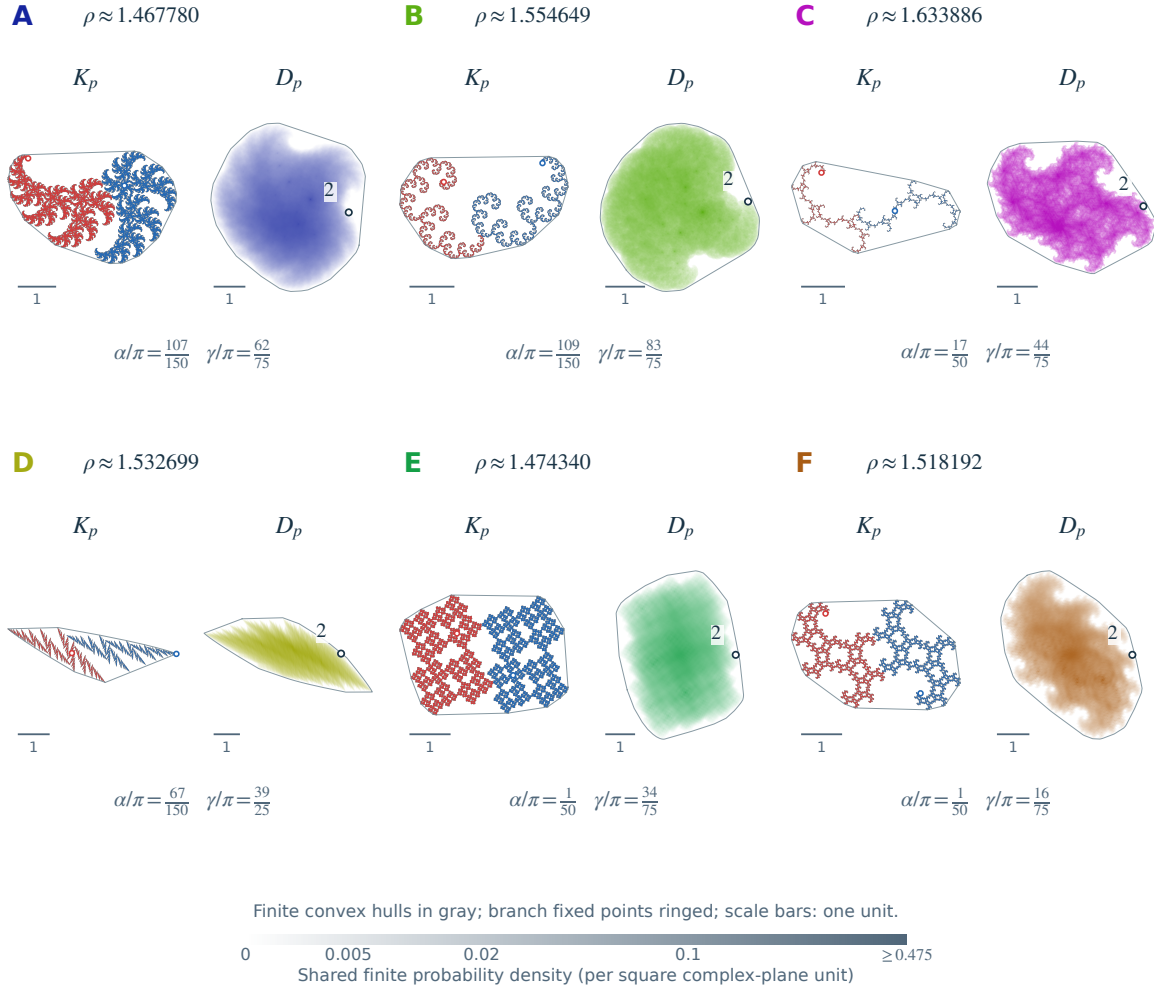


Figure 8: The six linked specimens. Red and blue show first-level pieces of K_p ; $D_p = aK_p - bK_p$ uses the corresponding atlas hue and a shared finite probability-density scale. Every attractor uses all length-22 addressed centres. Gray outlines are finite convex hulls, rings in K_p mark branch fixed points, and the ring in D_p marks 2. Scale bars have length one; panels are fitted independently. The images do not certify exact contact.

raster, display mesh, colour map and finite geometric depth. The explorer exports these independently of the image through JSON records and embeds parameter metadata in PNG files. The detailed laboratory also provides raw slice CSV and validated session records. These records allow a figure to be reproduced while preserving the distinction between a mathematical parameter, the computation performed there and its visual presentation.

11 Contact geometry and further directions

The atlas connects three levels of description. A parameter determines a compact attractor; its coding map records which symbolic points are identified; and its position within a prescribed contact family determines which of those identifications are forced. The exact examples show why these levels are useful together. The quarter-turn attractor has four different semilinear presentations with the same first-level pieces, while their agreement is established by explicit recodings. At the critical rectangle its first-level intersection is a segment, with different address multiplicities along that segment. On a whole marked binary zipper surface, by contrast, the stable set consists precisely of the embedded canonical curves. A single geometric contact label cannot describe all of these situations.

The triangle family makes the universal radial bounds concrete. Its elementary altitude dissection belongs to the classical geometry of self-similar tiles; related two-piece planar tiles are studied in [8]. Here the explicit normalization places such a family on the connectedness boundary at every contraction weight, and the separated perturbation proves sharpness in the full opposite–opposite chart. The opposite diagonal gives a second exact model: its circle and real antennae produce a discontinuous radial ceiling. Thus a smooth displayed sheet can be informative while missing genuine features of the parameter boundary.

Several precise questions follow from these constructions. Near a critical triangle or rectangle, which directions in the full coefficient space preserve contact, produce separation, or create positive-area overlap? The formulas for the exact families provide base parameters at which to study this local change. For a fixed eventually periodic pair, the contact equation is algebraic; one can ask when its regular branches organize nearby fixed-address fibres and when additional contact pairs change their local structure. These questions concern both the geometry of a fibre and the way fibres intersect.

The stable-set construction supplies a common language for the complex-tree families developed since [9, 10, 11] and for marked zippers. Calegari’s island theorem gives a disconnected embedded locus on one zipper surface [19]. It is natural to seek other prescribed contact families with disconnected stable sets, and to compare their forced quotients and excess-contact fibres. Passing from a slice to a larger family changes the forced relation, so any persistence statement should specify the family and coding relation under consideration. The exterior-coordinate comparison makes this distinction visible: the four selected stable slices are subsets of their marked zipper surfaces in the four-dimensional atlas, and colour retains their contraction weight after projection to three dimensions. Intersections in the projection can therefore arise from different weights; they do not themselves identify systems or contacts.

The companion atlas makes these questions accessible as reproducible experiments. Finite radial surveys suggest regions to investigate; exact contact relations provide positive examples; complete prefix covers and the contact-gap estimate provide open sets of certified separation. The next computational step is to adaptively refine those records around specified contact fibres and their intersections. This extends the atlas through explicit mathematical questions and checkable parameter data, while the visualization continues to support immediate movement between parameters and attractors.

A Reproducibility and exact computation records

The companion reproducibility archive separates the mathematical record from the current browser interface. It contains the figure-generation records, exact certificates, and public assets from which the scientific plates are recomposed; the arXiv source retains the files required to compile this article and the small exact certificate suite used below. The study uses all length-22 addressed centres for each attractor. Its preserved density rasters and atlas rendering are recomposed without asserting that the global radial survey has been rerun. The independent overlay script reconstructs each finite convex hull and checks the multipliers, branch fixed points and analytic tail formulas against the stored record. These are floating-point consistency checks.

The inspected companion snapshot is dated 18 September 2026. A source manifest records the original URL, byte size and SHA-256 digest of the inspected modules. The principal mathematical implementations are the family registry, the semilinear prefix renderer and the radial worker. The DD laboratory’s 43 173-ray data record reassembles to 5,913,659 bytes, with SHA-256 digest

5d833a368c0a43405d51ecd0e5a7a53c
ddd19a019142b9ed9f6f05e94a64af9.

The manifest distinguishes this frozen record from later changes to the public interface. The detailed laboratory’s source and data harness passed 73 checks; its grid, picking and session harness passed 3,381 enumerated checks. These results concern the inspected code and data, and are not frame-rate measurements.

A.1 A small exact benchmark suite

The reference computation uses rational complex coefficients and exact rational comparisons. Let $r_* < 1$ be a rational upper bound for both contraction ratios, verified by comparing their squared moduli with r_*^2 , and put $R_* = (1 - r_*)^{-1}$. For equal-length words u, v of length n , with opposite first symbols, write $c_u = S_u(0)$ and $c_v = S_v(0)$. The pair is excluded only if

$$|c_u - c_v|^2 > (2r_*^n R_*)^2. \quad (52)$$

The two cylinder images lie in the corresponding enclosing disks, so this strict inequality is a sufficient separation certificate.

The search starts with the first-piece pair, discards excluded pairs, and otherwise branches over all four choices of appended letters. Complete finite exhaustion certifies disconnection. A depth or pair limit leaves an unresolved frontier. The output records all excluded and unresolved leaves, together with the exact squared distances and bounds. The independent verifier reconstructs cylinder centres by evaluating the letters from right to left and checks both the inequalities and the complete four-child covering tree. It imports no producer code.

Table 2 gives the completed benchmark runs. The depth limit was 8 and the pair limit 20,000. Each row applies separately to every orientation type listed. All resulting records passed the standalone verifier.

For $a = b = 1/2$, the elementary interval identity $S_1([-2, 2]) = [-2, 0]$ and $S_2([-2, 2]) = [0, 2]$ proves that the attractor is $[-2, 2]$ in each orientation type. Thus the unresolved search output is consistent with a separate exact connectedness proof. The experiment checks the one-sided logic of exclusion; it does not implement a global classification of the atlas.

The `computation/` directory contains the producer `exact_exclusion.py` and the independent verifier `verify_certificate.py`. The seven JSON records and their summary are in

Coefficients $a = b$	Orientation types	Depth	Tests	Excluded	Output
1/4	(+, +)	1	1	1	Disconnected
2i/3	(+, +), (+, -), (-, -)	6	69	52	Disconnected
1/2	(+, +), (+, -), (-, -)	8	29	21	Unresolved

Table 2: Exact rational exclusion experiments. Depth counts the full cylinder word length, including its first symbol. Each $2i/3$ run has 17 split pairs and no surviving frontier; each $1/2$ run has seven split pairs and one surviving frontier pair. “Excluded” counts leaf pairs, not parameters.

computation/certificates/. These small examples establish a reproducible starting point for the computational method. Larger frontier claims require their own complete certificates or a proved positive criterion.

A.2 Parameter-neighbourhood proof objects

The companion’s version 2.0.0 rational producer and independent checker are included under `computation/neighbourhood/`. Their exact inputs specify the two multipliers as decimal or fraction strings and the two orientation bits. The producer uses individual rational outward bounds for word contractions; the checker reconstructs the affine maps with rational real matrices and checks the complete prefix cover. It then verifies the positive-gap neighbourhood from Lemma 9.2. Its 23-test suite includes independent exact-family benchmarks and rejection of incomplete or altered proof objects.

For example, the opposite–opposite parameter $a = b = 3(1 + i)/8$ is certified by four cover leaves of word length two. The exact checked gap exceeds $13/50$ and the parameter radius exceeds $1/300$. Consequently the complete product

$$|a' - 3(1 + i)/8| \leq 1/300, \quad |b' - 3(1 + i)/8| \leq 1/300$$

has disconnected attractors with first-piece gap at least $13/100$. The companion archive retains both the proof object and the independently checked record.

The two producers use the same mathematical separation criterion but different schemas and enclosure choices. Each proof object identifies its producer and is checked with the matching standalone checker. Rounding a trigonometric coefficient supplies an exact rational centre different from the intended trigonometric parameter; transferring a neighbourhood claim to the latter requires rigorous containment. The companion archive’s README gives the complete commands for the manuscript, figures and exact checks.

A.3 The four-family exterior comparison

Figures 4–6 transform positive certificates from a 512×256 dyadic grid on $[0, 1] \times [0, 1/2]$ in the endpoint q plane. Each cell has side $1/512$. The original endpoint-return calculation used a budget of 12 000 cylinder-pair visits and maximum word length 40. Success means that *every parameter* in the closed cell is embedded. A computation that does not finish gives no membership conclusion. The selected covers are DD10, DD11, DO10 and DO11. The DD10 record also includes cells obtained from DD01 by the exact reflected symmetry $q \mapsto 1 - \bar{q}$; the duplicate family is omitted from the figures. Complex conjugation supplies the reflected lower-half cover in the three-dimensional placement. The unpainted background carries no membership label. Exact OO stable-disk classifications are proved in the text and are not drawn in this four-family comparison.

The planar coordinate is $c = 1/q$; the three-dimensional position uses the canonical coefficients of Proposition 8.8, not just the endpoint multipliers. Both transformations are applied to the saved positive cells. The exterior panels retain the full 512×256 positive mask, clipped to $[0.96, 4.65] \times [-2.70, 0.13]$ in the c plane. The unbounded tail near $q = 0$ lies beyond this window. For the three-dimensional mesh, a 2×2 block is included only when all four original closed cells are certified; this thinning can omit positive cells but does not bridge uncertified gaps. The figure mesh approximates the curved images of those blocks. Exact cell coordinates and membership certificates are retained separately. The colour coordinate is w . The projection and image-generation records are retained in the companion archive under `figure_sources/four_families/`.

The interval implementation uses IEEE binary64 arithmetic with outward `nextafter` bounds, including contraction norms, invariant radii, cylinder centres and squared separation. Its recorded build uses GCC 13.3.0, with fast-math and fused contraction disabled. Gradual underflow and correctly rounded elementary operations, including square root, are assumed. An independent integer-polynomial checker verifies all fourteen common-return identities. A separate compilation reproduced the saved island-rectangle calculation; this is a replay of the same interval implementation, rather than a second numerical algorithm. The complete commands, exact parsed bounds, per-cell outcomes and file hashes are retained in the companion archive under `figure_sources/zippers/zipper_stability_run_manifest.json`.

For the curve examples, iterate the zipper operator from the identity map on $[0, 1]$ with equal partition intervals. Write $r = \max\{|q|, |1 - q|\}$. The first polygon differs from the identity by at most $|q - 1/2|$, so the depth- n polygon ζ_n satisfies

$$\|\zeta - \zeta_n\|_\infty \leq \frac{r^n |q - 1/2|}{1 - r}.$$

The figures use the three shared probes in (48). At q_A and q_B , depth 20 gives 1 048 577 vertices, with respective uniform bounds 1.824×10^{-6} and 1.493×10^{-4} .

At q_C the two contraction ratios differ substantially, so an adaptive prefix partition gives a better display for the same number of segments. First bound $E = \|\zeta - \text{id}\|_\infty$ using the exact depth-14 polygon and the tail estimate above: the maximum norm of its difference from the identity occurs at a polygon vertex. On a source cylinder with m occurrences of the first symbol and n of the second, the endpoint chord then has parametrized error at most $|q_C|^m |1 - q_C|^n E$. Subdivide until a rational upper bound for this product is at most $1/2000$. The leaves form an exact partition of $[0, 1]$, and all four curves have uniform error strictly less than 5×10^{-4} in exact arithmetic. The vertex counts and source depths are

Family	Vertices	Source depths
DD10	724 014	13–31
DD11	473 538	12–30
DO10	462 020	12–30
DO11	1 692 525	14–34

The plotted coordinates are evaluated in binary64; the analytic error bounds exclude this final floating-point evaluation and rasterization. Exact stopping decisions, source-partition checks and the resulting arrays are retained in the companion archive under `figure_sources/four_families/specimen_checks/`. Drawing a polygon without visible crossings is not used as a positive certificate. Ten positive interval records certify q_A, q_B in all four families and q_C in DD11 and DO11. Each test applies to a closed dyadic box of side 2^{-34} containing the stated exact rational parameter, with a budget of 2 000 000 cylinder-pair visits and maximum word length 100.

An additional exact calculation proves that the DO10 canonical curve at q_C is nonembedded. With letters 0, 1 denoting the first and second maps, respectively, take words $u = 00011100$ and $v = 10110111111$. Their closed parameter intervals are $[33/128, 67/256]$ and $[2303/4096, 9/16]$, which are disjoint. For $G(s, t) = F_u\zeta(s) - F_v\zeta(t)$ on $[0, 1]^2$, a cover by 46 cylinder enclosures keeps its boundary away from zero and certifies boundary degree -1 . Nonzero degree supplies a zero of G , hence an identification between the two disjoint parameter intervals. The coefficient arithmetic and strict disk-separation comparisons are rational; an independent checker reconstructs the enclosures and computes winding using a different ray. The proof object and replay record accompany the positive probe records in the companion archive under `figure_sources/four_families/specimen_checks/`. No conclusion about DD10 at q_C follows from its capped positive test.

Finally, the small DD11 square

$$\begin{aligned} 0.3409082591533661 &\leq \Re q \leq 0.3409092128276825, \\ 0.4348457604646683 &\leq \Im q \leq 0.4348467141389847 \end{aligned}$$

uses the exact dyadic endpoints recorded in hexadecimal in the run manifest; the displayed decimals are their round-trip representations. All $64^2 = 4096$ cells of side 2^{-26} pass, with a budget of 150 000 visits per cell and maximum word length 64. This proves embeddedness throughout the square, which contains Calegari's rational island parameter $L = 0.34090875 + 0.43484625i$. Its membership in a component separate from the real interval follows from the cited island theorem, not from this local calculation. The whole square has side 2^{-20} ; it is too small to see in the common-scale atlas and is not intended as a depiction of the island boundary.

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Data and code

The companion explorer is available at <https://complextrees.com/4D/>. The companion repository at <https://github.com/espigule/ComplexTrees> contains the figure-generation scripts, exact parameter records and finite disconnection certificates used in this article, together with an independent verifier. Section 10 gives the computational conventions, and the accompanying source record identifies the inspected explorer version and figure assets. All numerical images are labelled by their approximation method; the exact phase diagrams follow from the proofs in the text.

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